The Belt and Road Initiative and the Kazakhstan–China Trade Balance: Sources of the Post-2013 Swing and the Role of Strategic Mineral Exports, 2005–2024

Master’s thesis. Registered topic: BRI’s Impact on Kazakhstan–China Trade Balance: The Role of Strategic Mineral Exports

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Abstract

Between 2013, when the Belt and Road Initiative (BRI) was announced in Astana, and 2024, Kazakhstan's merchandise trade balance with China moved from a surplus of USD 3.5 billion to a deficit of USD 12.0 billion (China-reported data). The BRI is often credited, or blamed, for this swing. A bilateral deficit is not in itself an economic loss: imports can embody investment, integration and specialisation, and the BRI's main channels (lower trade costs, transit, investment) need not show up in the balance at all. The thesis therefore uses the balance as an accounting summary that organises the change in the whole merchandise relationship, exports and imports, and asks: **is the post-2013 change in Kazakhstan–China merchandise trade, summarised by the swing in the bilateral balance, distinctive to the relationship, and is it consistent with the BRI mechanisms that are observable in merchandise trade, in particular through strategic mineral exports?**

The thesis builds an HS-2 panel of Kazakhstan's exports to China and to the rest of the world for 2005–2024 from UN Comtrade, validated against China's mirror records and World Bank totals, and adds aggregate import data, China's trade with Russia and the Central Asian states, AidData records of Chinese official finance and dated connectivity milestones. It keeps three kinds of evidence distinct: accounting decompositions, comparison-adjusted (descriptive) components, and statistical tests on product-level data (PPML triple differences, event studies and trend breaks with Holm corrections, randomisation inference and alternative break years). No causal effect of the BRI is estimated.

Relative to Kazakhstan's trade with its other partners, most of the swing is specific to China: 52–92 per cent depending on base and end years, 71 per cent with averaged endpoints. Its largest part is a rise in imports from China after 2019. That rise was not specific to Kazakhstan: between 2019 and 2024 China's exports to each of Kazakhstan's five neighbours grew by a factor of 2.2–3.2, and to Kazakhstan by 2.2, the lowest of the six. On the export side, oil exports to China fell by more than to other markets, mainly between 2013 and 2014, contrary to the resource-security expectation; strategic mineral exports offset part of the swing in every specification; and agri-food exports reoriented towards China, matching the market-access mechanism in products, but without acceleration at any dated BRI milestone (2012, 2013, 2015, 2017). Chinese official finance peaked before 2013. Relative to Kazakhstan's other partners, bilateral trade turnover did not expand specifically with China before 2019.

The thesis concludes that the swing is real and mostly China-specific from Kazakhstan's side, that its largest component was shared by China's other neighbours, and that the merchandise-trade evidence does not show a change that both matches a BRI mechanism and starts or accelerates at a BRI milestone. This does not establish that the BRI had no effect on Kazakhstan: effects on transit, investment, productivity or welfare, region-wide effects and effects too small to detect are outside what the design can see.

Keywords: Belt and Road Initiative; Kazakhstan–China trade balance; strategic minerals; comparison-adjusted decomposition; PPML; mirror statistics; Central Asia

1. Introduction

1.1 The Phenomenon

China is one of Kazakhstan's two most important trading partners. In 2023 it supplied 27.4 per cent of Kazakhstan's imports and took 18.7 per cent of its exports (Bureau of National Statistics of the Republic of Kazakhstan, 2024). Kazakhstan is also where the BRI began: Xi Jinping proposed the Silk Road Economic Belt at Nazarbayev University in Astana on 7 September 2013 (China Daily, 2013). Connectivity between the two countries was improving before and after that date: a second rail border crossing, Altynkol–Khorgos, opened in December 2012 (Tengrinews, 2012); the Khorgos Gateway dry port began operating in 2015, with Chinese state companies taking a 49 per cent stake in 2017 (KTZ Express, 2025); Chinese policy banks financed road reconstruction in 2017–2018 (AidData, 2023); and the two governments signed protocols opening the Chinese market to Kazakh agricultural products (The Astana Times, 2023).

Over the same period Kazakhstan's merchandise trade balance with China swung by USD 15.5 billion: by China's reported data, from a surplus of USD 3.5 billion in 2013 to a deficit of USD 3.9 billion in 2019 and USD 12.0 billion in 2024 (Table 6.1). The composition of trade also changed: oil fell from 62 to 22 per cent of Kazakhstan's exports to China, metals and ores rose from 36 to 67 per cent, and imports from China more than doubled after 2019.

1.2 The Puzzle

The swing coincides with the BRI, and the BRI is the obvious bilateral explanation. But two other explanations predict the same before-and-after pattern. The swing may be the bilateral reflection of shocks that hit all of Kazakhstan's trade, such as the 2014–2016 collapse of oil and metal prices. Or it may be the Kazakh reflection of changes in China's trade with the whole region, such as the reorganisation of trade with Russia after 2022. Before-and-after comparisons cannot tell these apart. The puzzle is therefore: **Kazakhstan's merchandise trade with China changed sharply in the BRI era, but it is not known whether the change is distinctive to the Kazakhstan–China relationship at all, and if it is, whether it took the form the BRI's mechanisms imply.**

1.3 The Trade Balance and the Trade Relationship

The registered topic names the bilateral trade balance as the outcome. It is necessary to be precise about what the balance is and is not.

The balance is not the relationship. The BRI is usually discussed in terms of infrastructure, transport costs, trade facilitation, investment and trade expansion. None of these is the bilateral balance. Lower trade costs can raise both exports and imports, so the BRI could expand and deepen Kazakhstan–China trade while worsening, improving or leaving unchanged the bilateral balance. Imports of Chinese machinery and equipment can raise Kazakhstan's productive capacity; consumer imports can lower prices; goods can transit Kazakhstan to third countries. Machinery, electrical equipment and vehicles account for 47 per cent of the 2019–2024 rise in imports from China (Table 6.11). The BRI's best-documented channel for Kazakhstan, China–Europe rail transit (Mau & Seuren, 2023), earns service revenue that is not recorded in merchandise trade. A bilateral balance is also not a measure of gains from trade (Heckscher, 1919; Ricardo, 1817): Kazakhstan's total merchandise exports in 2024 (USD 82.5 billion) were close to their 2013 level (USD 84.7 billion) (World Bank, 2026). The thesis therefore does not treat the swing as a loss or the deficit as a problem.

What the balance is good for. First, it is an accounting identity: a swing of USD 15.5 billion must equal the sum of changes in specific exports and imports, so explaining it requires an account of the whole merchandise relationship, both sides and all product groups, rather than of one flow. The thesis uses it in that way: every result is a component of the swing, and bilateral trade turnover (exports plus imports), the outcome closest to the BRI trade-expansion literature, is reported alongside it (§6.2). Second, the balance with China is the indicator most often cited in Kazakhstan's public debate about the BRI, so establishing what moved it has direct policy relevance. Third, the registered topic asks about strategic mineral exports, and the balance is where their contribution can be set against the other components.

Scope of the conclusions. Conclusions are limited to Kazakhstan's merchandise trade with China: its balance, turnover and composition. The thesis does not assess the BRI's effects on transit services, investment, productivity or welfare.

1.4 Research Question

> **Central research question.** *Is the post-2013 change in Kazakhstan–China merchandise trade, summarised by the swing in the bilateral balance, distinctive to the Kazakhstan–China relationship, and is it consistent with the BRI mechanisms that are observable in merchandise trade, in particular through strategic mineral exports?*

"Distinctive" is assessed against two comparisons, which answer different questions, use different data and are not combined into a single number:

- **Relative to Kazakhstan's other partners.** Did trade with China change differently from Kazakhstan's trade with the rest of the world, product by product? This removes shocks common to Kazakhstan's trade, including world commodity prices.
- **Relative to China's other regional partners.** Did China's trade with Kazakhstan change differently from China's trade with Russia and the other Central Asian states? This removes shocks common to China's trade with the region. Because those partners are also BRI participants, it also removes any region-wide BRI effect.

"Consistent with a BRI mechanism" means that a change has the products, direction and timing that the mechanism predicts (Table 3.1). Timing is assessed against dated connectivity milestones (2012, 2013, 2015, 2017; Table 6.13b), not only against the 2013 announcement.

Four sub-questions organise the analysis:

- **SQ1.** Which flows and product groups account for the swing, and how much of it, and of bilateral turnover, is China-specific relative to Kazakhstan's other partners? (§6.1–6.2)
- **SQ2.** Which export groups, including strategic minerals, changed specifically towards or away from China, and did those changes begin or accelerate at BRI milestones? (§6.3–6.5)
- **SQ3.** Was the rise in imports from China specific to Kazakhstan relative to China's other regional partners, and when did it occur? (§6.6)
- **SQ4.** Do the timing and sectors of measurable BRI channels (Chinese finance, transport and border infrastructure, market access) match the components found under SQ1–SQ3? (§6.8)

The reliability of the bilateral statistics is a precondition, examined first (Chapter 4), because comparison-adjusted components are exactly what reporting discrepancies distort.

1.5 Why the BRI Is Central, and What Kind of Claim Is Made

The BRI is central for three reasons. First, it is the hypothesis the question is built around: the reason to ask whether the change is distinctive to the relationship is that the BRI is a bilateral and regional framework. Second, the BRI literature specifies mechanisms with observable implications for which products change and when (Chapters 2–3). Third, the conclusion is a statement about the BRI: which of its observable mechanisms the merchandise-trade evidence matches, which it contradicts, and where it is silent.

The thesis tests BRI *mechanisms* where they have observable implications in merchandise trade (products, direction, timing); it does not test the BRI as a whole. Three kinds of evidence are kept distinct:

1. **Accounting decompositions** (Table 6.2): identities that show which flows moved.
2. **Comparison-adjusted components** (Tables 6.3–6.4, 6.10b–c): how much a flow changed relative to a benchmark. They describe distinctiveness, not causes, and they are not a bound on any BRI effect: the BRI can change Kazakhstan's trade with other partners too, which moves the benchmark, and BRI effects inside a China-specific component can be offset by other China-specific changes.
3. **Statistical tests** on product-level data (Tables 6.7–6.9b, 6.12): whether China-specific export changes are distinguishable from noise and whether they began or accelerated at the predicted times.

Claims are graded at three levels: **A** (context: the BRI is the policy framework of the period), **B** (a distinctive change whose products, direction and timing match a BRI mechanism) and **C** (a causal effect). The design can reach A and B, never C. A test that cannot detect a change is *absence of evidence*; a change in the opposite direction or without the predicted timing is *evidence inconsistent with a specific mechanism's prediction*; neither is *evidence that the BRI had no effect*.

1.6 Aim, Objectives and Contribution

Aim. To determine what accounts for the post-2013 change in Kazakhstan–China merchandise trade, summarised by the balance, how far it is distinctive to the relationship, and whether its distinctive components are consistent with the BRI mechanisms observable in merchandise trade, including through strategic mineral exports.

Objectives.

1. To derive the BRI's expected mechanisms and their observable implications from the literature, including studies that find positive BRI trade effects (Chapters 2–3).
2. To build and validate a dataset in which distinctive components can be measured, reconciling Kazakh and Chinese statistics (Chapter 4).
3. To decompose the swing by flow and product group and compare it, and turnover, with Kazakhstan's other partners (§6.1–6.2).
4. To test which export groups changed specifically towards China, and when, against dated milestones (§6.3–6.5).
5. To compare the rise in imports with China's trade with Kazakhstan's neighbours (§6.6).

6. To compare BRI channels with these components and state what the evidence does and does not show for each mechanism (§6.8–6.9, Chapters 7–8).

Object and subject. The *object* is Kazakhstan–China bilateral merchandise trade. The *subject* is the composition and distinctiveness of the post-2013 change in that trade, summarised by the bilateral balance, and its relationship to BRI mechanisms and strategic mineral exports.

Contribution. The thesis replaces a before-and-after narrative with a two-sided comparison and a mechanism-by-mechanism assessment for one country pair. Its contribution is modest and specific: it shows which parts of the change in Kazakhstan–China merchandise trade were distinctive, from which side, and how each part relates to the observable predictions of BRI mechanisms (mechanism scorecard, §6.9). Supporting contributions are tests of the timing of the agri-food reorientation against several milestones and the documentation of a reporting break in Kazakhstan's 2020–2022 partner statistics.

Provisions submitted for defence.

1. Relative to Kazakhstan's other partners, most of the 2013–2024 swing is China-specific (52–92 per cent depending on base and end years; 71 per cent with averaged endpoints); its largest part is a rise in imports from China after 2019, and bilateral turnover did not expand specifically with China before 2019.
2. The post-2019 rise in imports from China was not specific to Kazakhstan: China's exports to each of Kazakhstan's five neighbours grew as fast or faster in 2019–2024.
3. On the export side, the China-specific fall in oil exports runs opposite to the resource-security expectation, strategic mineral exports offset part of the swing in every specification, and the agri-food reorientation matches the market-access mechanism in products but shows no acceleration at any dated milestone.
4. Kazakhstan's 2020–2022 partner statistics omit major metal exports and machinery imports; China's mirror data are needed to measure the relationship.

1.7 Relevance, Novelty and Significance

Relevance. China is one of Kazakhstan's two largest trading partners, and the bilateral deficit with China is widely cited in Kazakhstan as a measure of what the BRI has meant for the country. Whether that deficit reflects the BRI, the commodity cycle or regional developments matters for how trade, customs and statistical policy respond. The question is also timely: the largest part of the swing occurred in 2022–2024, during the reorganisation of trade in the region after 2022.

Scientific novelty. Within the literature reviewed here (Chapter 2), studies of the BRI and Kazakhstan–China trade either estimate average effects across many countries or describe one pair's trade before and after 2013. This thesis is, as far as the author has found, the first to (i) compare the change in Kazakhstan–China merchandise trade against two separate benchmarks, Kazakhstan's other partners and China's other regional partners; (ii) test the timing of China-specific export changes against dated connectivity milestones rather than the announcement alone; and (iii) document, with China's mirror data, the omission of major chapters from Kazakhstan's 2020–2022 partner statistics. The claim of novelty is limited to this combination for this country pair; each method is standard.

Theoretical significance. The thesis connects the BRI trade-cost and market-access literature, which predicts changes in trade volume and composition, with the policy debate about the bilateral balance, and shows what each mechanism implies for an outcome that the literature itself does not study.

Practical significance. The results are relevant to three audiences. For trade policy, they show that the post-2019 deficit is better treated as part of a regional import pattern than as a consequence of specific BRI projects. For agricultural export promotion, they identify agri-food as the one area where Kazakhstan's exports reoriented towards China, a reorientation that recorded Chinese finance did not target. For official statistics, they show that partner-level data for 2020–2022 need reconciliation with partner reports before they can support analysis of the relationship.

1.8 Structure

Chapter 2 reviews the literature, including evidence of positive BRI trade effects. Chapter 3 states what each mechanism predicts and what would count as support or contradiction. Chapter 4 builds and validates the data. Chapter 5 sets out the decompositions, comparisons, tests and safeguards. Chapter 6 answers the sub-questions and ends with a synthesis (Table 6.14) and a mechanism scorecard (§6.9). Chapter 7 interprets the results. Chapter 8 concludes.

2. Literature Review

2.1 Resource-Based Trade and the Comparative-Advantage Null

Standard trade theory predicts that a resource-abundant economy such as Kazakhstan exports resource-intensive goods and imports manufactures from a labour- and capital-abundant economy such as China (Heckscher, 1919; Ohlin, 1933; Ricardo, 1817). In that framework neither the bilateral balance nor the share of a particular partner has welfare significance in itself. What matters for this thesis is the empirical implication: if Kazakhstan's comparative advantage and world prices determine what it exports, the composition of its exports should change in the same way for every destination. Any change in the composition of exports to China would then reflect world prices and Kazakhstan's own supply, not anything specific to China. This is the null hypothesis against which BRI-related claims must be tested.

The commodity-dependence literature explains why that null is demanding. Commodity prices are volatile and have long cycles (Harvey et al., 2010), resource booms affect the real exchange rate and non-resource sectors (Corden & Neary, 1982), and the developmental consequences of resource wealth depend on institutions and policy (Frankel, 2010; Ploeg, 2011; Sachs & Warner, 1995). Export growth driven by resource volume can even reduce welfare when it worsens the terms of trade (Bhagwati, 1958). For Kazakhstan, the 2014–2016 oil price collapse alone could produce large changes in export composition, the bilateral balance and the exchange rate without any change in trade costs with China.

2.2 Trade Costs, Gravity and the BRI

Gravity models explain bilateral trade by economic size and bilateral trade costs relative to the trade costs each country faces with all partners (Anderson & Wincoop, 2003). Poisson pseudo-maximum-likelihood (PPML) estimation of such models is now standard because it handles zero flows and heteroskedasticity (Santos Silva & Tenreyro, 2006). The BRI literature applies this framework. De Soyres et al. (2019; 2019) estimate that BRI transport projects reduce shipping times and trade

costs, with gains concentrated along corridors. Baniya, Rocha and Ruta (2020) find that BRI-related improvements raise trade flows, more so when infrastructure is complemented by trade-policy reforms. Bird, Lebrand and Venables (2020) model Central Asia specifically and show that gains from improved connectivity are uneven across locations and sectors.

Two limits of this literature matter here. First, most estimates are simulations or cross-country averages; they do not test what happened in one country pair. Second, they predict *more trade*, not a particular balance or composition. The trade-cost channel has a compositional implication only through the next strand of theory: lower trade costs matter most for products that were previously not traded or were traded little.

2.3 Evidence of Positive BRI Trade Effects, and How It Relates to This Thesis

A substantial peer-reviewed literature finds that the BRI lowers trade costs and raises trade. Table 2.1 summarises the four studies most relevant here, with what each estimates.

Table 2.1. Selected studies of BRI trade effects and their comparability with this thesis

Study	Design	Outcome (dependent variable)	Main finding	Comparable to this thesis?
de Soyres et al. (2019), *International Economics*	Ex-ante; transport-network estimates of planned BRI projects	Shipment times and trade costs	Shipment times −1.7 to −3.2 per cent and trade costs −1.5 to −2.8 per cent in BRI economies; up to −11.9 and −10.2 per cent along some corridors	Mechanism only: predicts lower costs, not a bilateral balance
Baniya, Rocha and Ruta (2020), *Journal of Development Economics*	Ex-ante; estimated trade-time reductions fed into gravity and comparative-advantage models	Bilateral trade flows among 71 BRI economies	Trade flows up to +4.1 per cent; about three times more with trade reforms; larger gains for time-sensitive products	Predicts volume and composition (time-sensitive goods); not the balance
Foo, Lean and Salim (2020), *North American Journal of Economics and Finance*	Ex-post augmented gravity model, ASEAN and China, 2000–2016	Bilateral trade flows	Positive, statistically significant BRI (OBOR) dummy	Different region; total trade, not balance; dummy-based timing
Mau and Seuren (2023), *Review of World Economics*	Ex-post panel, timing of China–Europe rail connections, 1996–2018	European countries' exports to China, by industry	Rail connections raised European exports to China, including for indirectly connected countries	Transit corridor through Central Asia; outcome is third-country exports, not Kazakhstan's trade

All four entries were checked against the publisher's or RePEc's record of the abstract (Paper/evidence_audit.md). Effect sizes in the body of Mau and Seuren were not used. A Kazakhstan-specific simulation by Krulikowski and Riker (2024) is discussed in §2.6.

These studies and this thesis answer different questions, and their results are compatible with this thesis's findings. First, none of them uses a bilateral trade *balance* as the outcome; they measure trade costs, total trade or one side's exports. A BRI that raises both exports and imports can leave the balance better, worse or unchanged, so their positive findings neither predict nor contradict a particular direction of Kazakhstan's balance with China. Second, the ex-ante effects are a few per cent on average, smaller than this design can detect in one country pair's annual data (§6.7); a failure to detect them is absence of evidence. Third, the rail evidence concerns China–Europe transit, in which Kazakhstan is a corridor; transit earns Kazakhstan service revenue that merchandise trade data do not record. Fourth, the mechanisms these studies identify have specific implications for Kazakhstan's own trade with China (turnover, time-sensitive and non-traditional goods), and those implications are tested here (Chapter 3).

2.4 The Margins of Export Growth and Diversification

Models with heterogeneous firms imply that lower trade costs allow new, less productive firms and new products to enter a market (Melitz, 2003), so that bilateral trade responds to trade costs mainly through the *extensive margin* of new products and firms (Chaney, 2008). Across countries, larger exporters export more varieties (Hummels & Klenow, 2005). Diversification also follows a development pattern: production and exports diversify as income rises and re-concentrate only at high income levels (Cadot et al., 2011; Imbs & Wacziarg, 2003). These studies give the BRI trade-cost argument a specific, testable form: if BRI connectivity lowered the cost of reaching China, the gains should be largest for products that Kazakhstan exported little to China before, especially perishable or time-sensitive goods such as food, and the number of products exported to China should rise relative to other markets.

2.5 Chinese Finance and Resource Projects

A separate strand concerns Chinese official finance. China has become the largest official creditor to developing countries (Horn et al., 2021), and BRI lending raised debt-sustainability concerns in several corridor economies (Hurley et al., 2018), although claims of systematic “debt-trap diplomacy” are not supported by the evidence (Bräutigam, 2020). AidData’s Global Chinese Development Finance Dataset records Chinese official commitments project by project for 2000–2021 (AidData, 2023; Custer et al., 2023). For Kazakhstan, the relevant question is whether Chinese finance went into export capacity, for instance mines and processing, and whether that capacity was then oriented towards China. Jepson and Baldakova (2024) study whether Chinese state capital supported resource-based structural transformation in Bolivia and Kazakhstan, and Calabrese (2024) asks the same question for diversification in the Kyrgyz Republic. Global value chain research adds that suppliers in captive relationships often have limited scope for upgrading (Gereffi et al., 2005).

2.6 Kazakhstan, Central Asia and China

Area studies emphasise that the BRI in Central Asia is a broad label applied to projects, agreements and diplomatic initiatives of very different kinds (Cooley, 2012; Laruelle, 2018). Pomfret (2019) documents how Central Asian trade has been shaped by transport costs, trade policy and the Eurasian Economic Union, not only by China. Laruelle and Peyrouse (2012) and Grant (2020) show that the Chinese presence is received ambivalently in Kazakhstan, and Primiano and Kudebayeva (2025) find uneven views of the BRI among Kazakh students. Rolland (2017) and Swaine (2015) describe the strategic purposes China attaches to the initiative. This literature is a warning against treating “the BRI” as a single treatment with a single start date.

Quantitative studies of Kazakhstan–China trade itself are fewer. Huang, Yenina and Hossain (2021) apply Hummels–Klenow margin indices and intra-industry trade indices to China–Kazakhstan trade in 2002–2017 and describe a growing extensive margin of bilateral trade. Their indices are not benchmarked against Kazakhstan’s trade with other partners, so they cannot show whether that growth was specific to China; this thesis’s extensive-margin test (§6.7) adds that comparison. Krulikowski and Riker (2024), in a U.S. International Trade Commission working paper, combine a structural gravity model with a growth model and BRI membership dates, and simulate that the BRI raised Kazakhstan’s total exports by 10.7 per cent, its total imports by 15.4 per cent and China’s exports to Kazakhstan by 29.1 per cent in 2022 relative to a no-BRI counterfactual, with the largest gains among Central Asian countries. These are model-based average effects rather than

observed changes, and the paper is not peer-reviewed, but it is the closest existing estimate for this pair. It implies that the BRI may have widened Kazakhstan's bilateral deficit by raising imports more than exports, and that its effects were region-wide. Both implications bear directly on how this thesis's results should be read (§7.3).

2.7 Asymmetric Interdependence

Keohane and Nye (1977) distinguish *sensitivity*, how quickly changes in one economy affect another, from *vulnerability*, the cost of adjusting when a relationship is disrupted. Hirschman (1945) showed that trade can become a source of influence for the larger partner when the smaller partner's trade is concentrated on it and hard to redirect. This framework is frequently invoked for Kazakhstan–China relations. It does not, however, predict that trade concentration is harmful, or that it will be used as leverage; it predicts that the costs of disruption are distributed according to who can substitute more easily. Whether that is the case is an empirical question about both sides of the relationship, which most applications do not measure.

2.8 Research Gap

The literature supplies BRI mechanisms (trade costs and market access; resource finance), a null (comparative advantage and common shocks), and evidence that the BRI raises trade volumes on average. Among the studies reviewed here, the Kazakhstan-specific work either describes bilateral trade without a benchmark or simulates average BRI effects from membership dates; none asks whether the swing in Kazakhstan's balance with China is distinctive to the relationship, either relative to Kazakhstan's other partners or relative to China's other regional partners, or tests whether its distinctive components have the timing and composition BRI mechanisms imply. None checks the bilateral statistics on which such conclusions depend. This thesis addresses these gaps for one country pair; it does not estimate the BRI's effect in general, and it does not address transit services.

3. Theoretical Framework and Expectations

3.1 Framework

Following the structural gravity view of bilateral trade (Anderson & Wincoop, 2003), Kazakhstan's trade with China in product p and year t depends on:

1. **Factors common to Kazakhstan's trade with all partners:** world prices, Kazakhstan's output and income, its exchange rate and trade policy.
2. **Factors common to China's trade with the region:** China's export supply and import demand, regional routing, and regional shocks such as the post-2022 reorganisation of trade with Russia.
3. **Persistent features of the Kazakhstan–China relationship:** distance, pipelines and rail links that existed before 2013, long-standing contracts.
4. **Changes specific to the Kazakhstan–China relationship over time:** BRI-era infrastructure at the Kazakhstan–China border, bilateral market-access agreements, Chinese finance in Kazakhstan, and other bilateral changes.

The BRI can operate through (2), (3) and (4): it is both a bilateral and a regional initiative. The comparison against Kazakhstan's other partners removes (1); the comparison against China's other regional partners removes (2), including any

region-wide BRI effect. Neither comparison isolates the BRI, which is why mechanism predictions about products and timing are needed.

3.2 Mechanisms and What Would Count as Evidence

The BRI-related mechanisms relevant to Kazakhstan–China merchandise trade are:

M1. Trade-cost reduction and connectivity (Baniya et al., 2020; Bird et al., 2020; Soyres, Mulabdic, Murray, et al., 2019). New and upgraded rail, road and border infrastructure lowers the cost of trade between the two countries. Expected: an expansion of bilateral trade turnover beyond Kazakhstan's trade with others, after the infrastructure becomes operational.

M2. Time-sensitive and non-traditional goods, and market access (Baniya et al., 2020; Chaney, 2008; Melitz, 2003). Lower costs and phytosanitary access raise trade most where it was previously small and for time-sensitive goods such as food. Expected: a China-specific rise in agri-food and other non-traditional exports, beginning or accelerating after dated milestones; more product chapters exported to China.

M3. Resource finance and security (Horn et al., 2021; Jepson & Baldakova, 2024). Chinese finance for hydrocarbon and mining projects raises Kazakhstan's China-oriented exports of energy and strategic minerals. Expected: China-specific rises in energy and mineral exports after the relevant finance and project completion.

M4. Investment-linked imports. Chinese-financed projects and Chinese investment raise imports of Chinese machinery and equipment. Expected: a Kazakhstan-specific rise in capital-goods imports from China timed with project activity.

M5. Transit. Kazakhstan earns revenue from China–Europe transit (Mau & Seuren, 2023). Expected: growth in transit volumes and service receipts. This is not observable in merchandise trade and is not tested.

N0. Common shocks and comparative advantage (Harvey et al., 2010; Heckscher, 1919; Ohlin, 1933). Kazakhstan–China trade moves with Kazakhstan's trade with others and with China's trade with the region.

Table 3.1. Observable implications

Mechanism	Observation that would support it	Observation inconsistent with its prediction	Uninformative
M1 Trade costs, connectivity	China-specific rise in turnover after operational milestones (2012–2019)	No China-specific turnover expansion while infrastructure became operational	A rise shared by all of China's neighbours (could be region-wide BRI or other regional shocks)
M2 Time-sensitive goods, market access	China-specific rise in agri-food and other non-traditional exports that starts or accelerates after a milestone; positive extensive margin	A rise that began earlier and does not accelerate at any milestone	A change smaller than the test can detect
M3 Resource finance	China-specific rise in energy and mineral exports after Chinese finance	China-specific fall; gains unrelated in time to finance	Gains whose timing depends on the base year
M4 Investment-linked imports	Kazakhstan-specific rise in capital-goods imports from China timed with projects	Rise in imports shared equally by neighbours without comparable projects	Aggregate imports without project-level data
M5 Transit	Transit volumes and service receipts		All merchandise-trade evidence
N0 Common shocks	Comparison-adjusted components near zero	Large components in either comparison	

A component is classed as **consistent with a BRI mechanism (level B)** only if it is distinctive, has the predicted sign and products, and begins or accelerates after a dated milestone. **Inconsistent with a mechanism's prediction** means the

evidence contradicts that specific prediction; it does not mean the BRI had no effect.

3.3 How the Expectations Were Formed (Disclosure)

The general structure (a common-shocks null against connectivity and resource-finance mechanisms) was planned before the product-level data were collected. The product-group predictions were written after the author had inspected aggregate Kazakhstan–China data (in the earlier version of this thesis) and the descriptive shares by product group in the new panel (Table 6.6). The PPML tests were run after that inspection. The trend-break test, the comparison-adjusted decomposition, the comparison with China's regional partners, the robustness grid over base and end years, the independent re-estimation, and, in the final validation, the alternative break years (Table 6.9b), the comparators excluding Kyrgyzstan and Uzbekistan alone, the partner-by-partner growth comparison and the turnover comparison were added in later rounds of analysis, after earlier results had been reviewed. These elements are therefore exploratory or confirmatory-after-exposure, not pre-registered. §5.7 describes the safeguards used.

3.4 Where Asymmetric Interdependence Fits

Asymmetric interdependence (Hirschman, 1945; Keohane & Nye, 1977) is used in Chapter 7 to interpret the results, not as a hypothesis tested by them. The decompositions measure how trade changed; they do not measure bargaining power. Chapter 7 uses two-sided indicators of dependence without assuming that concentration of trade with China implies Chinese leverage.

4. Data

4.1 Sources

All trade data come from the UN Comtrade database, retrieved through its public preview API on 31 October 2026 (United Nations Statistics Division, 2026a). Every value used is transcribed in a raw text file in `Collected_Raw_Data/raw_downloads/`, with the query URL, reporter, partner, flow, units and retrieval date in the file header (Appendix A). Table 4.1 lists the series.

Table 4.1. Data series

Series	Reporter → partner	Flow	Valuation	HS level	Years	File
Kazakhstan's exports to China	Kazakhstan (398) → China (156)	Exports	FOB	HS-2	2005–2024	`comtrade_hs2_kaz_exports_to_china.txt`
Kazakhstan's exports to the world	Kazakhstan → World (0)	Exports	FOB	HS-2	2005–2024 except 2020, 2022	`comtrade_hs2_kaz_exports_to_world.txt` (+ gap-fill file)
China's imports from Kazakhstan (mirror)	China → Kazakhstan	Imports	CIF	HS-2	2005–2024 except 2006	`comtrade_hs2_chn_imports_from_kaz.txt`

Series	Reporter → partner	Flow	Valuation	HS level	Years	File
China's exports to Kazakhstan (mirror of Kazakhstan's imports)	China → Kazakhstan	Exports	FOB	HS-2	2013, 2019, 2024	`comtrade_hs2_chn_exports_to_kaz.txt`
Kazakhstan's exports to Russia, HS 84, 85, 87	Kazakhstan → Russia (643)	Exports	FOB	HS-2	2019, 2023	`comtrade_hs2_kaz_exports_to_russia_selected.txt`
Merchandise exports, total	World Bank WDI	Exports		Total	2005–2024	`wdi_kaz_merch_exports.csv` (World Bank, 2026)
Kazakhstan's total imports from World and from China	Kazakhstan → World, China	Imports	CIF	Total	2011–2014, 2019, 2023, 2024	`comtrade_totals_imports_and_china_exports.txt`
China's total exports to and imports from Kazakhstan, Russia, Kyrgyzstan, Uzbekistan, Tajikistan, Turkmenistan, World	China → partners	Exports (FOB) and imports (CIF)		Total	2011–2014, 2019, 2023, 2024	`comtrade_china_totals_by_partner.txt`
China's imports from World, selected chapters	China → World	Imports	CIF	HS-2	2024	`comtrade_hs2_chn_imports_from_world_2024_selected.txt`
Chinese official finance commitments	AidData GCDF 3.0	Commitments	Constant 2021 USD	Project	2000–2021	`aiddata_gcdf_v3.xlsx` (AidData, 2023)

Values are in current US dollars unless stated. HS chapters are those reported (HS 2002 to HS 2022 vintages); chapter boundaries are stable across these vintages at the two-digit level.

Data were retrieved year by year because the public API returns one period per call. Where the retrieval tool truncated a long response, the missing chapters were identified by comparison with neighbouring years and re-queried explicitly; these are recorded in a separate gap-fill file. One response (Kazakhstan to World, 2022) was rejected because its tail contained duplicated values and omitted the two largest chapters; Kazakhstan-to-World data for 2020 could not be retrieved. As §4.4 shows, 2020–2022 would have been excluded in any case.

4.2 Product Groups

The 97 HS chapters are grouped into four groups. **Energy** is HS 27 (crude oil, gas, coal and products). **Metals and ores** are HS 25, 26, 28, 71, 72 and 74–81: salt and minerals, ores and concentrates, inorganic chemicals including uranium compounds (HS 2844), precious metals, iron and steel including ferroalloys, and base metals (copper, nickel, aluminium, lead, zinc, tin and others). **Agri-food** is HS 01–24. **Other goods** are all remaining chapters, mainly chemicals, plastics, textiles, metal articles, machinery and vehicles. The "strategic minerals" of the registered topic (uranium, copper, ores and ferroalloys)

fall inside the metals and ores group and are reported separately where relevant (§6.2). Energy is a single chapter, which matters for inference (§5.7).

4.3 Validation Against Independent Totals

Table 4.2 compares the sum of Kazakhstan's reported HS-2 exports to the world with the World Bank's merchandise export totals, and Kazakhstan's reported exports to China with China's reported imports from Kazakhstan.

Table 4.2. Validation of the HS-2 data (USD bn)

Year	KZ exports to world, sum of chapters	WDI merchandise exports	Ratio, %	KZ-reported exports to China	China-reported imports from KZ	Mirror ratio
2005	27.85	27.85	100.0	2.42	2.91	1.20
2008	71.17	71.17	100.0	7.68	7.73	1.01
2010	57.24	59.97	95.5	10.12	11.11	1.10
2011	88.10	84.34	104.5	16.29	15.35	0.94
2012	92.28	86.45	106.7	16.48	14.68	0.89
2013	84.66	84.70	100.0	14.37	16.05	1.12
2016	36.74	36.69	100.1	4.22	4.81	1.14
2019	58.07	57.31	101.3	8.00	8.86	1.11
2020	n/a	46.45		6.84	10.37	**1.52**
2021	53.07	60.63	**87.5**	6.85	11.29	**1.65**
2022	n/a	84.59		10.46	14.82	**1.42**
2023	78.74	78.53	100.3	14.76	16.36	1.11
2024	81.61	82.47	99.0	14.90	15.91	1.07

Source: UN Comtrade (United Nations Statistics Division, 2026a); World Bank (World Bank, 2026). Author's calculations (Script 60, `hs2_validation.csv`; all years in the file). Mirror ratio = China-reported imports (CIF) / Kazakhstan-reported exports (FOB).

In most years the chapter sums match the World Bank totals within about 1 per cent, which confirms that the transcribed data are complete. The exceptions are informative. In 2011–2012 Comtrade's totals exceed the World Bank's by 4–7 per cent; this is a difference between the two publishers' totals that affects both destinations equally and is absorbed by the fixed effects in Chapter 5. In 2010 the chapter sum is 4.5 per cent below the WDI total; the cause could not be identified. In 2021 it is 12.5 per cent below, which is explained by the suppression described in §4.4.

The mirror ratio of 1.0–1.2 in normal years is what one expects when one side reports CIF and the other FOB. In 2020–2022 it jumps to 1.4–1.65.

4.4 A Reporting Break: Suppressed Chapters in 2020–2022

The jump in the mirror ratio has a precise cause. In Kazakhstan's own partner-level data for 2020–2022, exports to China of copper (HS 74), aluminium (HS 76) and zinc (HS 79) are absent, although China reports importing USD 1.8–2.5 billion of Kazakh copper in each of those years (Table 4.3). In 2021, Kazakhstan's reported copper exports to the *world* are only USD 0.11 billion, against USD 2.7 billion in 2019 and USD 3.2 billion in 2023. The same pattern was found earlier for imports: Kazakhstan's partner-level records for 2020–2022 contain no machinery (HS 84), electrical equipment (HS 85) or

vehicle (HS 87) imports from China, and its reported imports from China fall by about 70 per cent in 2020 while China's reported exports to Kazakhstan fall by 8 per cent (Appendix B, Table B.2).

Table 4.3. Copper, aluminium and zinc exports to China by reporter (USD million)

Year	Copper: KZ-reported	Copper: China-reported	Zinc: KZ-reported	Zinc: China-reported
2017	1,110	1,230	333	340
2018	1,365	1,470	167	558
2019	1,567	1,743	563	623
2020	not reported	1,834	not reported	450
2021	not reported	2,451	not reported	586
2022	not reported	2,388	not reported	33
2023	1,912	2,160	270	315
2024	2,695	2,742	327	413

Source: UN Comtrade (United Nations Statistics Division, 2026a). Author's calculations (Script 61, `mirror_suppression.csv`).

The pattern is consistent with confidentiality suppression of chapters dominated by few firms, but the reason cannot be confirmed from the data. Two consequences follow. First, Kazakhstan-reported product-level data for 2020–2022 are excluded from all comparisons of China with the rest of the world. Second, the bilateral balance is computed from China's records, which show no such gaps (§6.1). The main export results are re-estimated with China's mirror data for the China side (§6.7, Table 6.7).

4.5 The Persistent Import Gap

China's reported exports to Kazakhstan have long exceeded Kazakhstan's reported imports from China: by a factor of 1.5 to 2.5 in 2013–2019, before the 2020–2022 break (Appendix B, Table B.2). Valuation cannot explain it: Kazakhstan records imports CIF and China records exports FOB, so Kazakhstan's figures should be the higher ones. It may reflect goods declared differently at the border, transit through the Eurasian Economic Union customs territory, or under-declaration; the data cannot distinguish these. For this reason the thesis does not use Kazakhstan's import figures to measure the balance. The main balance figures in Chapter 6 are China-reported (China's imports from Kazakhstan minus China's exports to Kazakhstan), and are therefore not comparable in level with figures published by Kazakhstan's Bureau of National Statistics.

4.6 Chinese Official Finance

AidData's GCDF 3.0 records 155 Chinese official finance projects in Kazakhstan with commitments in 2000–2021 that AidData recommends for aggregate analysis (AidData, 2023; Custer et al., 2023). Amounts are in constant 2021 US dollars. The data record commitments, not disbursements, and they cover official finance only, not private Chinese investment. Syndicated loans appear as one record per lender. These limitations mean the data can show where and when Chinese state finance was committed, but not how much was spent or what it produced.

4.7 Summary of Discrepancies and Substitutions

Table 4.4. Data discrepancies, their likely causes and their treatment

Discrepancy	Evidence	Most likely cause	Treatment	Remaining limitation
KZ exports of HS 74, 76, 79 absent in 2020–2022	China reports USD 1.8–2.5 bn of copper imports per year; KZ copper exports to the World only USD 0.11 bn in 2021 (Table 4.3)	Missing partner records, consistent with confidentiality suppression; not valuation (the gap is total, not proportional)	Years excluded from KZ-reported product comparisons; China legs China-reported	Cause cannot be confirmed
KZ imports of HS 84, 85, 87 absent in 2020–2022	No monthly records in these chapters in 2020–2022, present in all other years (Table B.2)	Missing partner records	KZ-reported imports not used for 2020–2022; China's export data used	As above
KZ imports from China 1.5–2.5 times below China's exports in normal years	Table B.2; ratio 1.84 in 2024	Not valuation: Kazakhstan's CIF figures should exceed China's FOB figures, so the CIF/FOB difference widens rather than explains the gap. The cause is unresolved (candidates: under-declaration, attribution by country of origin versus consignment, cross-border shuttle trade, transit within the Eurasian Economic Union)	Levels and the balance from China's data; KZ-reported robustness for comparisons (Table 6.3)	The level of the balance depends on the reporter; comparison-adjusted shares do not (92 per cent with either)
KZ exports to China vs China's imports from KZ, ratio 0.89–1.28 outside 2020–2022	Table 4.2	CIF/FOB valuation and timing	Either source; China's for comparisons with China-reported imports	Small year-to-year noise
Comtrade KZ export totals above WDI in 2011–2012 (4–7 per cent), below in 2010 (4.5 per cent)	Table 4.2	Differences between publishers' totals	Affects China and RoW alike; absorbed by chapter-by-year effects; base-period averages include these years	Cause not identified
Truncated API responses	Missing chapters in some years	Retrieval tool length limit	Missing chapters re-queried; totals validated against WDI	None detected after gap-fill
KZ-to-World 2022 response with duplicated values	Identical values for different chapters; HS 27 and 28 absent	Retrieval error	Rejected; year not used	2022 unavailable
Mixed reporters in the comparison against KZ's other partners	China legs China-reported, RoW growth KZ-reported	Necessary because KZ partner data have gaps	Robustness with KZ-reported China legs	Coverage differences between reporters cannot be fully excluded

Throughout, HS chapters are as reported in each year; the two-digit chapter boundaries are stable across the HS vintages in the data, so no concordance is needed at this level.

5. Empirical Strategy

The strategy follows the two parts of the central question. §5.1 decomposes the swing and compares it with Kazakhstan's other partners (SQ1). §5.2–5.4 test the export components (SQ2), and §5.5 compares imports with China's other regional partners (SQ3). §5.6 states the identification limits, and §5.7 the safeguards against multiple testing, exploratory analysis, specification choice and missing data.

5.1 Accounting Decomposition and Comparison Against Kazakhstan's Other Partners (SQ1)

Accounting decomposition. With all terms China-reported, the change in Kazakhstan's balance with China between periods t_0 and t_1 is

$$\Delta B = \sum_g \Delta X_g^{CN} - \Delta M^{CN}.$$

For 2013–2024: $\Delta B = (15.91 - 27.95) - (16.05 - 12.55) = -15.55$ (USD bn). This is an identity.

Comparison against Kazakhstan's other partners. For each export chapter *p*, the benchmark assumes that exports to China grow at the rate of Kazakhstan's exports of *p* to the rest of the world (RoW):

$$X^{B}_{p,t_1} = X^{CN}_{p,t_0} \cdot \left(X^{RoW}_{p,t_1} / X^{RoW}_{p,t_0} \right)$$

The **China-specific export component** is $\sum_p \left(X^{CN}_{p,t_1} - X^{B}_{p,t_1} \right)$ and the **common component** is $\sum_p \left(X^{B}_{p,t_1} - X^{CN}_{p,t_0} \right)$; they sum to the actual change. Where a chapter's RoW base is below USD 1 million, its product group's RoW growth is used. For imports, the same calculation is done on totals: imports from China are benchmarked against the growth of Kazakhstan's imports from all other suppliers (Kazakhstan's imports from the world minus those from China, Kazakhstan-reported). The balance's China-specific component is the export component minus the import component, and its share is that component divided by the actual change in the balance (reported only when both have the same sign). The China-specific component of turnover is the export component plus the import component.

Data sources: the China legs are China-reported (China's imports from Kazakhstan, CIF; China's exports to Kazakhstan, FOB), because Kazakhstan's partner data have gaps (§4.4–4.5); the RoW growth rates are Kazakhstan-reported. A robustness version uses Kazakhstan-reported China legs throughout.

What the components are and are not. The benchmark is a description of how trade with China would have moved had it moved like trade with everyone else; it is not an estimate of what would have happened without the BRI. The China-specific component therefore measures distinctiveness, not causation. It is not an upper bound on a BRI effect, for two reasons: the BRI can change Kazakhstan's trade with other partners too, which moves the benchmark; and BRI effects within the China-specific component can be offset by other China-specific changes. Shift-share components are not additive across sub-periods, and they depend on the base and end years, so they are reported for a grid of alternatives: base years 2012, 2013, 2014 and the 2011–2013 average; end years 2019, 2023, 2024 and the 2023–2024 average (Script 67).

5.2 Descriptive Indicators

China's share of Kazakhstan's exports is computed by product group from Kazakhstan's data. Concentration is measured by the Herfindahl–Hirschman index across HS-2 chapters, with and without HS 27, and by the number of chapters exported above USD 1 million.

5.3 Triple-Difference PPML for the Export Side (SQ2)

The shift-share split shows which groups have large China-specific components; the PPML model tests whether they are statistically distinguishable from zero and robust. It compares, within each chapter and year, exports to China with exports to RoW:

$$X_{pdt} = \exp\left(\alpha_{pt} + \gamma_{pd} + \sum_g \beta_g \left(\mathrm{China}_d \times \mathrm{Post}_t \times G_{pg} \right) \right) \varepsilon_{pdt},$$

where *p* is the chapter, $d \in$ {China, RoW}, Post = 1 from 2014 and *G* indicates the four product groups.

- **Chapter-by-year effects** α_{pt} absorb everything common to all destinations: world prices, Kazakhstan's production, exchange rate and export policy. They are the regression counterpart of the "common component".
- **Chapter-by-destination effects** γ_{pd} absorb the pre-2014 orientation of each chapter towards China.

- **β_g** is the post-2013 change in the China/RoW ratio for group *g*; the ratio changes by a factor of exp(β).

Estimation is by PPML (Santos Silva & Tenreyro, 2006) in `pyfixest`, with standard errors clustered by chapter. Chapter–destination pairs that are always zero are dropped. Main sample: 2005–2019 and 2023–2024 (17 years, 97 chapters, 3,160 observations).

5.4 Timing: Event Study and Trend Break (SQ2)

β_g compares post-2013 levels with the *pre-2013 average*, so any upward trend yields a positive β whether or not it began with the BRI. Timing is therefore tested directly: an event study replaces Post with year dummies (base 2013), with a joint test of the pre-2013 coefficients; and a trend-break model estimates the China-specific trend before 2013 (θ_g) and its change afterwards (δ_g):

$$X_{pdt} = \exp\left(\alpha_{pt} + \gamma_{pd} + \sum_g \left[\theta_g \,\mathrm{China}_d G_{pg}\,(\,t-2013\,) + \delta_g \,\mathrm{China}_d G_{pg}\,(\,t-2013\,)\,\mathrm{Post}_t\right]\right)\varepsilon_{pdt}.$$

Consistency with M2 requires $\delta_g > 0$ (acceleration after the break). The relevant test is therefore one-sided. Because the 2013 announcement is not the date at which infrastructure became operational, the model is also re-estimated with the break at the dated operational and financing milestones of 2012, 2015 and 2017 (Table 6.13b; exploratory).

5.5 Comparison Against China's Other Regional Partners, and the Extensive Margin (SQ3)

The second comparison uses only China's own reports, so each flow has one reporter and one valuation. The benchmark assumes that China's exports to Kazakhstan grow at the rate of China's exports to a comparator *C*:

$$M^B_{KZ,t_1} = M_{KZ,t_0} \cdot G_C, \qquad \text{Kazakhstan-specific import change} = M_{KZ,t_1} - M^B_{KZ,t_1}.$$

Comparators are the world; Russia; Central Asia excluding Kazakhstan (Kyrgyzstan, Uzbekistan, Tajikistan, Turkmenistan); the same excluding Kyrgyzstan, a known re-export hub; Uzbekistan alone; and Central Asia plus Russia. Growth factors for each partner separately are also reported, which normalises for the large differences in partner size. G_C is computed in two ways: value-weighted (the comparator group's total in t_1 divided by its total in t_0) and as the median of the individual partners' growth factors, which limits the influence of any one partner. Results are reported with and without Russia and for the same grid of base and end years as §5.1.

The same comparison is *not* used for Kazakhstan's exports. China's imports from Russia and Turkmenistan are dominated by oil and gas, whose prices and post-2022 volumes move for reasons unrelated to Kazakhstan, so the regional benchmark for exports is not comparable (Script 67 computes it for completeness). Export distinctiveness is assessed product by product against Kazakhstan's other markets (§5.1, §5.3).

The two comparisons are not nested and are not applied sequentially: they use different reporters, different benchmarks and answer different questions. They are reported side by side.

The extensive margin on the export side is tested by a linear probability model of exporting more than USD 1 million, with the same fixed effects as §5.3.

5.6 Identification Limits

The design identifies China-specific change, not its cause. A rise in China's demand for Kazakh sunflower oil, a phytosanitary protocol, a faster border crossing and a Chinese-financed road would produce the same China-specific component. This is why the China-specific component is treated as a description of distinctiveness, not as an estimate or bound of a BRI effect, and why consistency with a BRI mechanism (level B) is the strongest claim made.

5.7 Safeguards: Exploratory Analysis, Multiple Testing and Missing Data

Limiting the search. All export tests use four product groups fixed by HS chapter boundaries before estimation (§4.2): energy, metals and ores, agri-food and other. The analysis does not search across individual chapters for significant results. Energy is a single chapter and has no valid standard error, so each family of tests has three inferential coefficients.

Multiple testing. Holm–Bonferroni adjustments are applied within each family: the three group coefficients of the main triple difference, and the three trend-break coefficients. Robustness variants (Table 6.7) are not counted as separate discoveries; a result is used only if it holds in the main specification after adjustment.

Randomisation inference. For agri-food, the group that drives the export-side conclusions, 300 random sets of 24 chapters are drawn from the 96 non-energy chapters and the China × Post and trend-break coefficients are re-estimated. This asks whether agri-food is unusual among Kazakhstan's export chapters, which guards against both small-cluster problems and the possibility that any set of chapters would show a similar change.

Exploratory elements. As disclosed in §3.3, the group predictions, the trend-break test, the comparison-adjusted decomposition, the neighbour comparison and the robustness grid were added after inspecting data. The decompositions are descriptive and involve no test. For the tests, the safeguards above apply, and the thesis reports which conclusions would change without them.

Specification grids instead of single choices. Every comparison-adjusted figure that enters a conclusion is reported over a grid of four base periods and four end periods (Tables 6.3b, 6.4, 6.10b) and, for imports, six comparator specifications. A conclusion is stated only if it holds across the grid; otherwise the range is reported.

Independent re-estimation. With chapter-by-year fixed effects and two destinations, the Poisson model in §5.3 is equivalent to a binomial (logit) model of China's share in each chapter-year, weighted by the chapter-year total. The main coefficients were re-estimated this way with a different library (`statsmodels`) as a check on the `pyfixest` estimates (§6.7).

Power. For the trend-break test, the minimum detectable change in the China-specific trend (one-sided 5 per cent test, 80 per cent power) is reported, so that a non-significant result is not mistaken for evidence of no change.

Missing data. Missingness has four sources, handled differently:

1. *Confidentiality-type suppression in Kazakhstan's 2020–2022 partner data* (§4.4). This is not random: it removes metals and machinery. These years are excluded from all Kazakhstan-reported product comparisons and from decomposition endpoints; China's mirror data, which show no suppression, are used for the China legs, and the main PPML is re-estimated with them.
2. *Retrieval failures*: Kazakhstan-to-World HS-2 data for 2020 and 2022 and China-reported imports for 2006. These are failures of the retrieval process, unrelated to trade values; they reduce the sample but are not expected to bias it. All three

years are outside the decomposition endpoints.

3. *Truncated API responses*: missing chapters were re-queried; completeness was verified against World Bank totals (§4.3).
4. *Limited import detail*: China's exports to Kazakhstan by chapter were collected for 2013, 2019 and 2024 only, and imports from RoW only as totals. The import side is therefore analysed at aggregate level, and import results are presented as descriptive.

6. Results

The chapter answers the sub-questions in order: the composition of the swing and its China-specific part (§6.1–6.2, SQ1), its export side (§6.3–6.5, SQ2), its import side (§6.6, SQ3), the robustness of the inferential results (§6.7), and the BRI channels (§6.8, SQ4). §6.9 maps every result to the central question.

6.1 The Balance and Its Gross Components

Table 6.1. Kazakhstan's trade with China, China-reported data (USD bn)

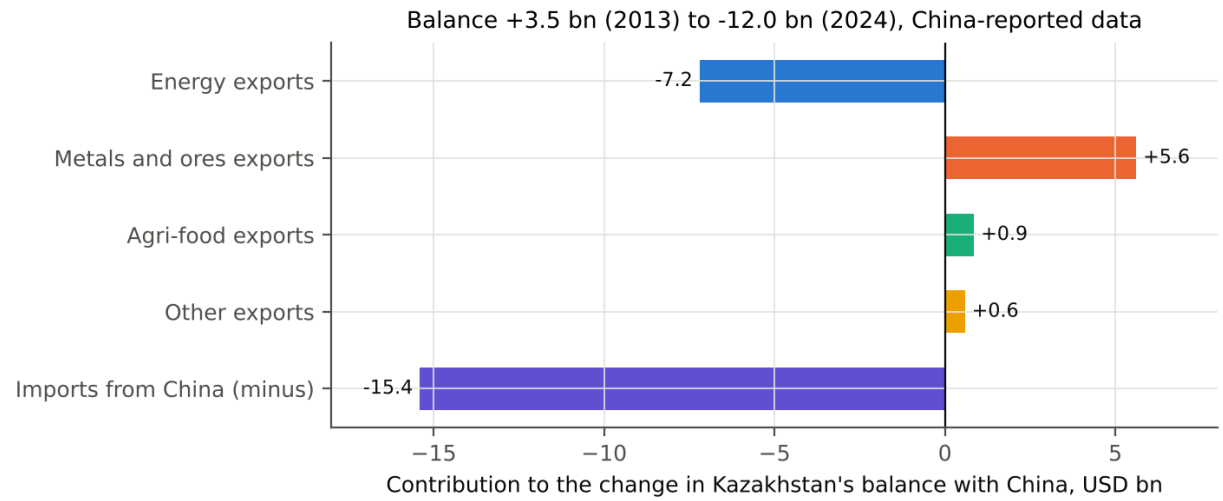
	2013	2019	2024
Kazakhstan's exports to China (China's imports, CIF)	16.05	8.86	15.91
Kazakhstan's imports from China (China's exports, FOB)	12.55	12.73	27.95
Balance for Kazakhstan	+3.51	−3.87	−12.04

Source: UN Comtrade, China as reporter (United Nations Statistics Division, 2026a). Author's calculations (Script 61, `trade_balance_decomposition.csv`).

Table 6.2. Gross decomposition of the change in the balance (USD bn)

Component	2013→2019	2019→2024	2013→2024
Energy exports	−7.56	+0.36	−7.20
Metals and ores exports	+0.06	+5.56	+5.62
Agri-food exports	+0.30	+0.56	+0.86
Other exports	+0.02	+0.57	+0.59
Imports from China (enters with a minus sign)	+0.18	+15.23	+15.41
Change in balance	**−7.37**	**−8.18**	**−15.55**

Source: as Table 6.1.



Source: UN Comtrade (HS-2, public API); author's calculations.

Figure 6.1. Gross decomposition of the change in Kazakhstan's balance with China, 2013–2024

The swing has two phases. In 2013–2019 the balance fell by USD 7.4 billion while energy exports to China fell by USD 7.6 billion; imports were flat. In 2019–2024 the balance fell by a further USD 8.2 billion because imports rose by USD 15.2 billion, more than offsetting a USD 7.0 billion rise in exports, most of it metals and ores. These are gross changes; they do not yet say whether anything was specific to China.

6.2 Is the Swing China-Specific Relative to Kazakhstan's Other Partners? (SQ1)

Table 6.3. Comparison against Kazakhstan's other partners, 2013 to 2024 (USD bn)

Period	Flow	Actual change	Common	China-specific
2013→2019	Exports	−7.19	−3.74	−3.44
	Imports	+0.18	−2.33	+2.52
	Balance	**−7.37**	**−1.41**	**−5.96**
2019→2024	Exports	+7.05	+2.81	+4.24
	Imports	+15.23	+4.53	+10.70
	Balance	**−8.18**	**−1.72**	**−6.46**
2013→2024	Exports	−0.14	+0.06	−0.20
	Imports	+15.41	+1.30	+14.11
	Balance	**−15.55**	**−1.24**	**−14.31**
Kazakhstan-reported China legs				
2013→2024	**Balance**	**−6.27**	**−0.52**	**−5.74**

§5.1. Export China legs: China-reported imports from Kazakhstan by HS-2 chapter; RoW growth: Kazakhstan-reported exports to World minus China, by chapter. Import China leg: China-reported exports to Kazakhstan; RoW growth: Kazakhstan-reported imports from World minus China (×0.81 in 2013–2019, ×1.36 in 2019–2024, ×1.10 in 2013–2024). Components are not additive across sub-periods. Source: Script 65, `balance_common_vs_china_specific.csv`.

For 2013–2024, USD 14.31 billion of the USD 15.55 billion swing is China-specific (92 per cent: 14.31 ÷ 15.55). With Kazakhstan's own data the swing is smaller in level (USD 6.27 billion, because Kazakhstan records much lower imports from China, §4.5), but the China-specific share is the same (5.74 ÷ 6.27 = 92 per cent).

The 92 per cent figure is the *highest* value in the robustness grid, because 2013 was the peak year of oil exports to China. Table 6.3b reports all base and end combinations.

Table 6.3b. China-specific share of the swing under alternative base and end years (China-reported China legs)

Base End	2019	2023	2024	2023–24 average
2011–13 average	58% (−4.76 of −8.19)	64% (−8.05 of −12.66)	77% (−12.55 of −16.36)	**71% (−10.30 of −14.51)**
2012	52% (−3.90 of −7.54)	57% (−6.82 of −12.01)	71% (−11.12 of −15.72)	65% (−8.97 of −13.87)
2013	81% (−5.96 of −7.37)	82% (−9.68 of −11.84)	92% (−14.31 of −15.55)	88% (−11.99 of −13.70)
2014	71% (−0.64 of −0.90)	56% (−3.01 of −5.37)	83% (−7.58 of −9.07)	73% (−5.29 of −7.22)

China-specific balance component and actual change in USD bn in brackets. Source: Script 67, `robust_kz_partner_grid.csv`*.*

Three results hold in every cell of the grid. (i) Most of the swing is China-specific: between 52 and 92 per cent, and 71 per cent with averaged endpoints, which are least sensitive to single-year spikes. (ii) The China-specific *import* component is small to 2019 (+0.4 to +3.0 billion) and large by 2023–2024 (+8.1 to +14.7 billion). (iii) The China-specific *export* component is negative to 2019 under every base (−0.2 to −3.4 billion). The China-specific export component to 2024, in contrast, depends on the base year: −0.2 billion from 2013, but +2.0 to +3.7 billion from 2011–2013, 2012 or 2014. Any statement that the export side "nets to about zero" is specific to the 2013 base and is not made.

Turnover. The trade-expansion literature (§2.3) concerns trade volume rather than the balance. Applying the same comparison to bilateral turnover (exports plus imports), the China-specific component of turnover is between −0.9 and +2.0 billion to 2019, under every base year, and between +12.2 and +18.3 billion to 2023–2024, of which the import component is +8.1 to +14.7 billion (`robust_kz_partner_grid.csv`). Relative to Kazakhstan's other partners, then, Kazakhstan–China trade did not expand specifically while most BRI connectivity projects were being built and opened (2013–2019); it expanded after 2019, mainly through imports.

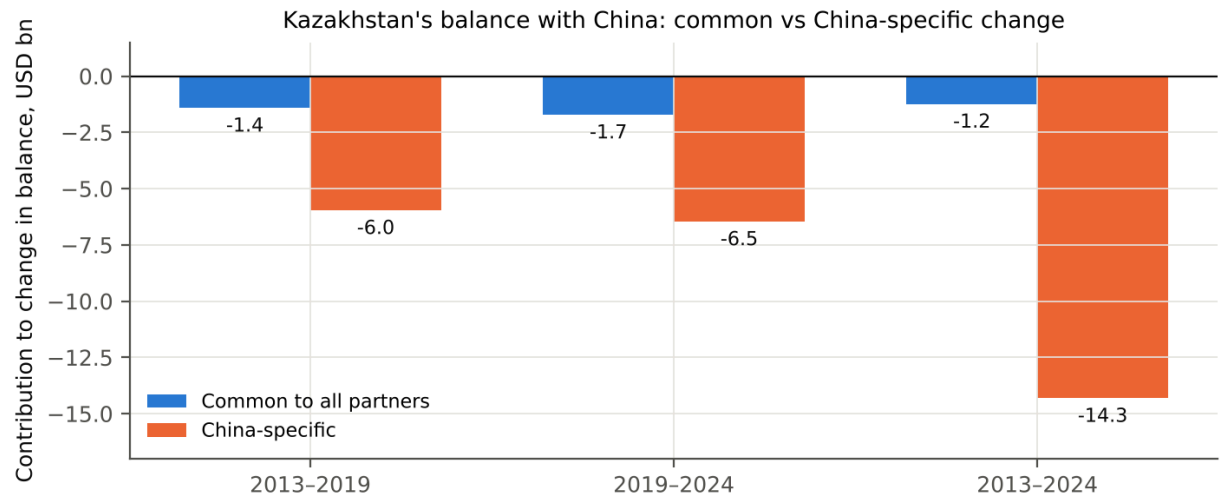
Table 6.4. China-specific export component by product group (USD bn)

Group	2013→2019	2019→2024	2013→2024	Range across the grid, end 2023–2024
Energy	−3.93	−0.16	−4.90	−4.9 to +2.4 (negative from every pre-2014 base)
Metals and ores	+0.20	+3.80	+3.36	+1.6 to +6.6
of which strategic minerals (HS 26, 28, 72, 74, 79)	+0.38	+3.70	+3.42	+1.8 to +6.7
Agri-food	+0.25	+0.13	+0.82	+0.7 to +0.9
Other	+0.03	+0.47	+0.52	+0.3 to +0.5
Total exports	**−3.44**	**+4.24**	**−0.20**	−0.2 to +5.1

China-reported China legs. With Kazakhstan-reported China legs, 2013→2024: energy −3.61, metals and ores +2.97, agri-food +0.89, other −0.08. Range: base 2011–13 average, 2012, 2013 or 2014; end 2023, 2024 or 2023–24 average. Sources: Script 65, `export_components_by_group.csv`*,* `strategic_minerals_components.csv`*; Script 67,* `robust_export_groups_grid.csv`*.*

By product group:

1. **Energy.** Oil exports to China fell by more than oil exports to other markets. From any base before 2014 the China-specific component is negative at every end year (−1.2 to −4.9 billion). From a 2014 base it is small and changes sign (−0.7 to +2.4 billion): the China-specific fall is concentrated in the move from 2013 to 2014, and its size at the end depends on the volatile oil exports of 2023–2024.
2. **Strategic minerals.** Their China-specific component is positive under every base and end year (+1.8 to +6.7 billion by 2023–2024), so they offset part of the swing in every specification. How much of this had happened by 2019 depends on the base: +0.4 billion from 2013, +1.7 billion from the 2011–2013 average, +2.4 billion from 2012.
3. **Agri-food and other goods.** Small positive China-specific components in every specification.



Source: UN Comtrade (HS-2, public API); author's calculations. Shift-share decomposition, China-reported China legs (Script 65).

Figure 6.2. Common and China-specific components of the change in Kazakhstan's balance with China

The remaining sections ask whether these components are distinguishable from noise (export side), whether the import component is distinctive from China's side, and whether either matches BRI mechanisms.

6.3 Export Composition and China's Share

Table 6.5. Kazakhstan's exports to China by product group (Kazakhstan-reported)

Year	Total, USD bn	Energy	Metals and ores	Agri-food	Other
2005	2.42	20.2%	66.7%	0.2%	12.9%
2008	7.68	57.6%	36.4%	0.0%	6.0%
2013	14.37	62.0%	36.4%	0.5%	1.1%
2016	4.22	24.0%	70.9%	3.1%	2.0%
2019	8.00	35.3%	58.3%	4.6%	1.8%
2023	14.76	36.0%	54.8%	6.5%	2.7%
2024	14.90	22.0%	67.0%	6.6%	4.4%

Source: UN Comtrade, Kazakhstan as reporter. Author's calculations (Script 61, `composition_kz_exports_to_china.csv`, which also gives the China-reported shares; the two sources differ by up to 10 percentage points in individual years, mainly because of CIF/FOB valuation and timing, but show the same direction of change).

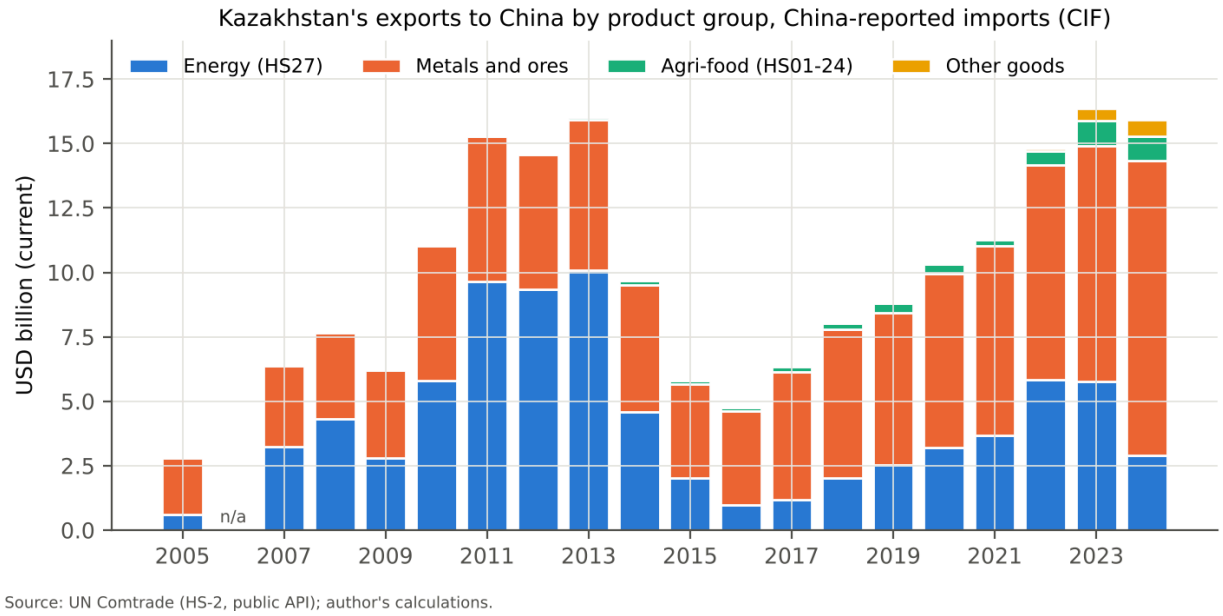


Figure 6.3. Kazakhstan's exports to China by product group, China-reported, 2005–2024

Table 6.6. China's share of Kazakhstan's exports by product group, % (Kazakhstan-reported)

Year	Energy	Metals and ores	Agri-food	Other	All goods
2005	2.5	24.5	0.7	30.1	8.7
2008	9.0	16.7	0.1	18.3	10.8
2010	13.5	33.7	0.8	6.8	17.7
2012	14.1	33.0	1.9	2.3	17.9
2013	13.8	34.8	2.5	6.8	17.0
2016	4.5	28.2	6.1	4.9	11.5
2019	7.4	33.4	11.1	6.0	13.8
2023	11.4	39.9	18.0	6.0	18.7
2024	7.1	44.2	19.4	8.5	18.3

Source: as Table 6.5 (`china_share_by_group.csv`, all years). 2020–2022 omitted (§4.4).

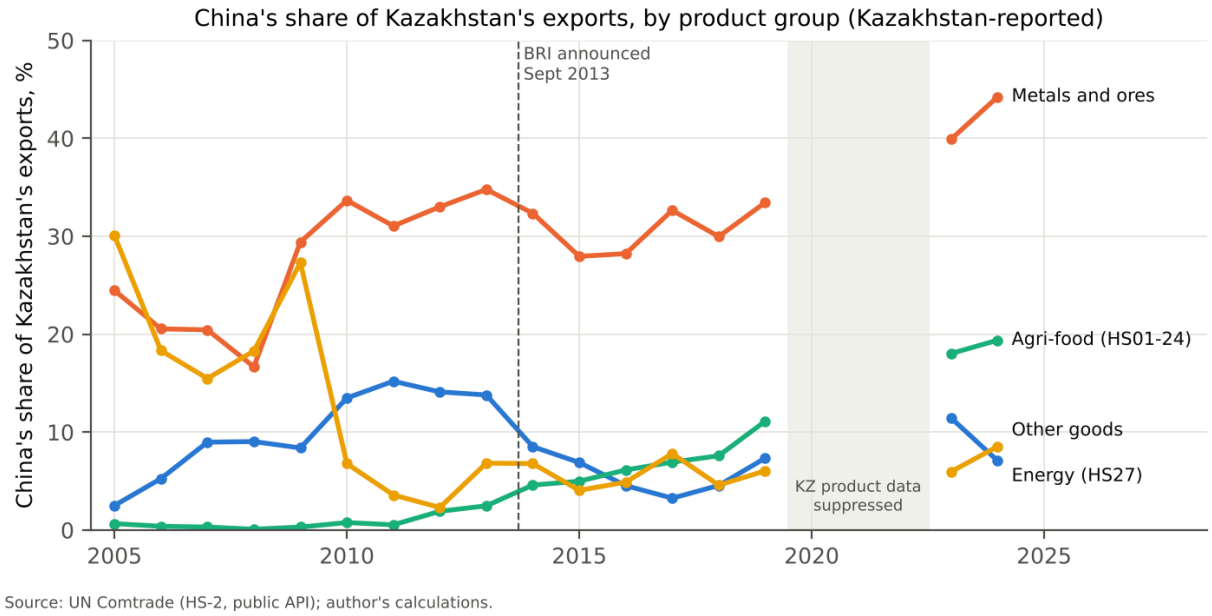


Figure 6.4. China's share of Kazakhstan's exports by product group

China's share of Kazakhstan's energy exports rose from 2.5 to 14–15 per cent between 2005 and 2011, then fell back to 3–7.4 per cent in 2016–2019. China's share of metals and ores rose from 17 to 34 per cent between 2008 and 2010, stayed at 28–35 per cent until 2019, and reached 40–44 per cent only in 2023–2024. China's share of agri-food rose steadily, from below 1 per cent before 2011 to 2.5 per cent in 2013, 11 per cent in 2019 and 19 per cent in 2024.

Strategic minerals. China's share of Kazakhstan's ore exports (HS 26) was 44 per cent in 2013 and 41 per cent in 2019, then 80 per cent in 2024; its share of copper (HS 74) rose from 56 to 64 per cent between 2013 and 2024; its share of inorganic chemicals including uranium compounds (HS 28) fell from 42 per cent in 2013 to 23 per cent in 2019 and was 32 per cent in 2024 (Script 61). The largest single movement is the post-2019 shift of ores towards China; how much of the strategic-mineral gain in Table 6.4 had occurred by 2019 depends on the base year.

Concentration. The HHI of Kazakhstan's exports to China fell from 0.415 in 2013 to 0.173 in 2024, but almost entirely because oil fell: excluding HS 27 it was 0.212 in 2013 and 0.205 in 2024 (`diversification_indices.csv`). Non-resource goods rose from 1.5 to 11.0 per cent of exports to China, against 6.8 to 16.7 per cent of exports to the rest of the world.

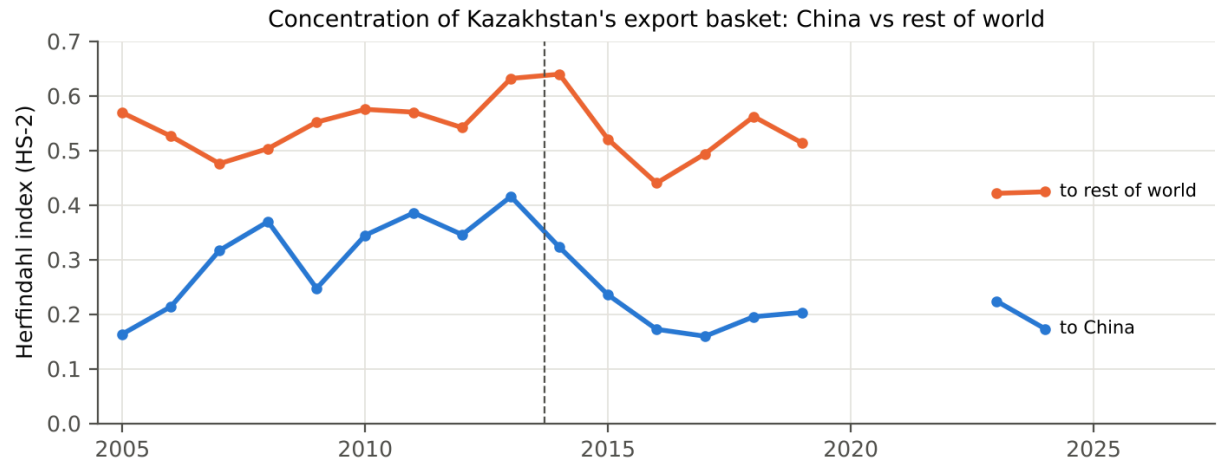


Figure 6.5. Concentration (HHI) of Kazakhstan's exports to China and to the rest of the world

6.4 Which Export Changes Were Statistically China-Specific? (SQ2)

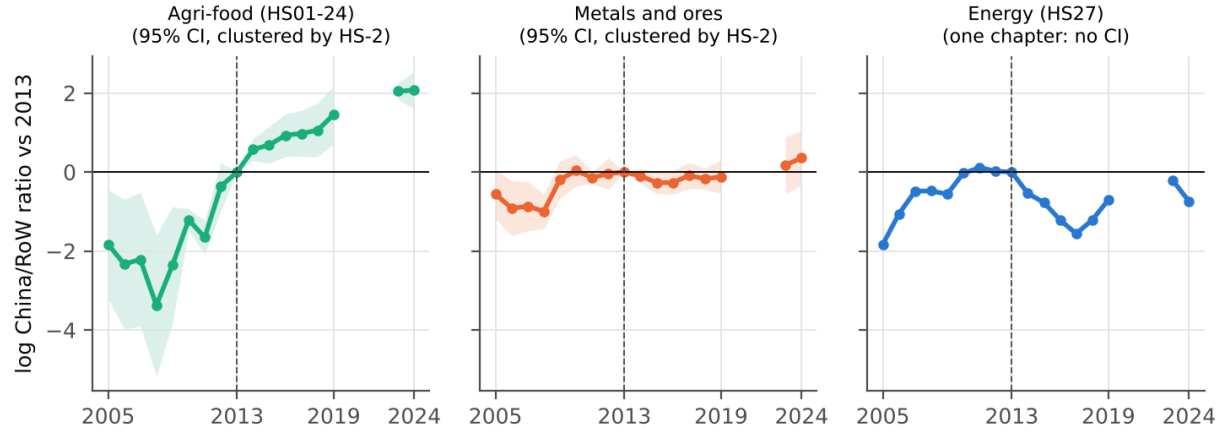
Table 6.7. PPML triple difference: change in the China/RoW export ratio after 2013, by product group

Sample	N	Energy	Metals and ores	Agri-food	Other
(1) Main: KZ-reported, 2005–19 and 2023–24	3,160	−0.528 [n/a]	0.329 (0.074)***	2.456 (0.213)***	0.634 (0.522)
(2) Mirror: China-reported imports for China leg	2,754	−0.659 [n/a]	0.436 (0.112)***	2.307 (0.292)***	1.144 (0.628)*
(3) 2005–2019 only	2,784	−0.669 [n/a]	0.154 (0.096)	1.986 (0.250)***	0.194 (0.379)
(4) Post from 2015	3,160	−0.539 [n/a]	0.319 (0.100)***	2.120 (0.206)***	0.508 (0.544)
(5) Post from 2017	3,160	−0.446 [n/a]	0.381 (0.149)**	1.827 (0.184)***	0.594 (0.540)
(6) Excluding HS 71	3,126	−0.528 [n/a]	0.328 (0.074)***	2.456 (0.213)***	0.634 (0.522)
(7) Chapters with mean > USD 1m	2,532	−0.528 [n/a]	0.329 (0.074)***	2.462 (0.214)***	0.640 (0.523)
(8) Placebo: 2005–2013, false post = 2010	1,617	0.722 [n/a]	0.663 (0.162)***	2.607 (0.876)***	−0.228 (0.733)

*Coefficients on China × Post × group. Chapter-by-year and chapter-by-destination fixed effects. Standard errors clustered by HS-2 chapter in parentheses (unadjusted; Holm-adjusted values in Table 6.12). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Energy is a single chapter, so no standard error can be computed [n/a]. Source: Script 62, `ppml_main.csv`.*

The regression confirms the shift-share picture. Relative to the rest of the world, the China ratio fell by 41 per cent for energy, rose by 39 per cent for metals and ores and by a factor of 11.7 for agri-food (row 1). Other goods show no significant change. The agri-food result holds in every variant. The metals result depends on 2023–2024 (row 3). The placebo (row 8), which pretends the BRI began in 2010 and uses only pre-BRI years, finds large "effects" for agri-food and metals: both were already changing before 2013.

6.5 Did the Export Changes Begin with the BRI? (SQ2, timing)



Source: UN Comtrade (HS-2, public API); author's calculations. PPML with product-year and product-destination fixed effects (Script 62).

Figure 6.6. Event study: China/RoW export ratio by product group relative to 2013

Table 6.8. Event-study coefficients (log China/RoW ratio relative to 2013)

Year	Agri-food	Metals and ores	Energy
2005	−1.83 (0.70)	−0.56 (0.32)	−1.83
2008	−3.37 (0.91)	−0.99 (0.22)	−0.48
2010	−1.21 (0.14)	0.04 (0.20)	−0.03
2011	−1.65 (0.21)	−0.14 (0.07)	0.11
2012	−0.36 (0.31)	−0.04 (0.20)	0.03
2013	0	0	0
2014	0.57 (0.14)	−0.10 (0.10)	−0.54
2016	0.93 (0.27)	−0.27 (0.15)	−1.21
2017	0.98 (0.30)	−0.09 (0.17)	−1.56
2019	1.45 (0.37)	−0.12 (0.21)	−0.70
2023	2.05 (0.11)	0.18 (0.36)	−0.21
2024	2.07 (0.24)	0.36 (0.35)	−0.74

PPML with chapter-by-year and chapter-by-destination fixed effects; SE clustered by chapter in parentheses (none for energy). All years in `ppml_event_study.csv` *(Script 62).*

Table 6.9. Trend-break test: China-specific trend before and after 2013 (log points per year)

Group	Chapters	Pre-2013 trend θ	Change after 2013 δ (95% CI)	One-sided *p* for acceleration (δ > 0)
Agri-food	24	0.538	−0.368 (−0.676 to −0.059)	0.99
Metals and ores	12	0.095	−0.070 (−0.194 to 0.053)	0.87
Other	60	−0.017	0.107 (−0.163 to 0.376)	0.22
Energy	1	0.068	−0.134 [n/a]	n/a

PPML with chapter-by-year and chapter-by-destination fixed effects; SE clustered by chapter. Source: Script 62, `ppml_trend_break.csv`*; Script 66,* `multiple_testing.csv`*.*

Energy. China's orientation of Kazakhstan's energy exports rose until 2011 and fell sharply from 2014 (−1.56 log points by 2017, relative to 2013) before partly recovering. The China-specific oil decline therefore began with the 2014 price collapse and ran *opposite* to the resource-security mechanism M3. No inference is possible with one chapter.

Metals and ores, including strategic minerals. The China orientation rose before 2010, was flat or slightly negative between 2013 and 2019, and turned positive only in 2023–2024. There is no detectable acceleration after 2013 (one-sided p = 0.87), although the test could miss a change smaller than about 0.16 log points per year (§6.7). In the regression the rise in China orientation appears only in 2023–2024; in the decomposition, the part of the strategic-mineral gain that occurred by 2019 depends on the base year (§6.2).

Agri-food. The China/RoW ratio rose steeply before the BRI: from −3.37 in 2008 to −0.36 in 2012, relative to 2013 (pre-2013 coefficients jointly different from zero, Wald $\chi^2(8) = 664$, $p < 0.001$). After 2013 it kept rising but more slowly: the China-specific trend fell from 0.54 to 0.17 log points per year. There is no evidence of the acceleration that M2 requires: the one-sided p-value for $\delta > 0$ is 0.99, and the whole unadjusted 95 per cent interval for δ lies below zero, so a non-negative trend change is outside it (with three tests in the family, the corresponding Holm-adjusted p-value for a two-sided test is 0.059, §6.7). Whether the trend significantly *slowed* is less certain (§6.7). In absolute terms, most of the growth came after 2013, from USD 70 million in 2013 to USD 980 million in 2024, led by vegetable oils (HS 15), fodder (HS 23), cereals (HS 10) and oilseeds (HS 12) (`agrifood_chapters.csv`). Both facts hold: the dollar expansion happened in the BRI period, but the China-specific growth rate did not rise when the BRI began.

Table 6.9b. Trend change at alternative break years (exploratory)

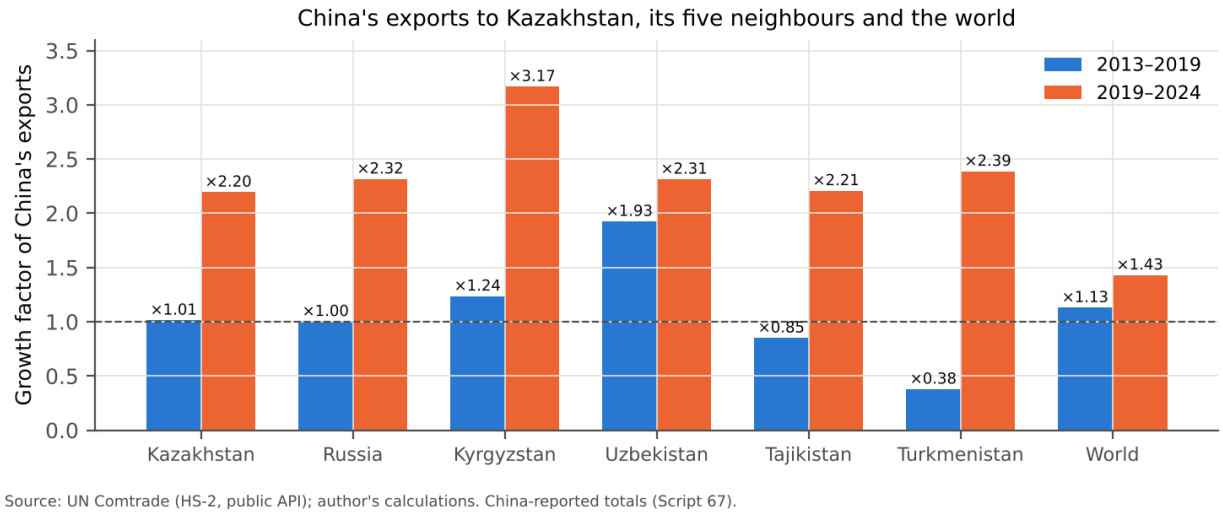
Break year (milestone)	Agri-food: δ (95% CI)	One-sided *p*, acceleration	Metals and ores: δ (95% CI)	One-sided *p*	Other: δ (95% CI)	One-sided *p*
2012 (Altynkol–Khorgos rail crossing)	−0.395 (−0.837 to 0.047)	0.96	−0.090 (−0.215 to 0.034)	0.92	0.138 (−0.141 to 0.416)	0.17
2013 (BRI announcement; main test)	−0.368 (−0.676 to −0.059)	0.99	−0.070 (−0.194 to 0.053)	0.87	0.107 (−0.163 to 0.376)	0.22
2015 (Khorgos Gateway dry port)	−0.255 (−0.460 to −0.050)	0.99	−0.033 (−0.158 to 0.092)	0.70	0.114 (−0.117 to 0.345)	0.17
2017 (Khorgos stake; road loans)	−0.169 (−0.350 to 0.011)	0.97	−0.011 (−0.140 to 0.119)	0.56	0.141 (−0.079 to 0.360)	0.10

Same PPML trend-break model as Table 6.9 with the break moved to each year; unadjusted 95% intervals. Break years were chosen after the 2013 result was seen, so this table is exploratory. Source: Script 69, `trend_break_alternative_years.csv`.

Moving the break to the dated connectivity milestones does not change the conclusion for agri-food or metals: at no milestone does the China-specific trend accelerate. For other goods the point estimates are positive at every milestone but imprecise (one-sided p 0.10–0.22): absence of evidence, not evidence against.

6.6 Was the Import Surge Specific to Kazakhstan? (SQ3)

The China-specific import component is the largest part of the swing (USD 14.1 billion over 2013–2024, Table 6.3). Its timing and Kazakhstan-specificity can be checked against China’s trade with neighbouring countries.



Source: UN Comtrade (HS-2, public API); author's calculations. China-reported totals (Script 67).

Figure 6.7. Growth of China's exports to Kazakhstan, its five neighbours and the world

The rise came after 2019 and mainly in 2022–2024: China's reported exports to Kazakhstan were USD 12.7 billion in 2019, 11.7 billion in 2020, 13.8 billion in 2021, 16.4 billion in 2022, 24.7 billion in 2023 and 28.0 billion in 2024 (Table B.2 and Table 6.1). For 2013–2019, whether there was any China-specific import rise relative to Kazakhstan's other suppliers depends on the data source: with China's export figures it is +USD 2.5 billion, with Kazakhstan's own import figures it is zero (Script 65), because the gap between the two sources widened in these years (§4.5). Tables 6.10b and 6.10c compare China's exports to Kazakhstan with its exports to Kazakhstan's neighbours.

Table 6.10b. Imports from China: Kazakhstan-specific change relative to China's exports to comparators (USD bn, China-reported)

Comparator (weighting)	2013→2019	2013→2024	2011–13 avg → 2023–24 avg	Range, end 2019	Range, end 2023–2024
Russia	+0.14	−1.21	−1.93	+0.14 to +0.95	−3.37 to +0.66
Central Asia excl. Kazakhstan (value-weighted)	−2.91	−14.38	−13.85	−2.91 to −1.51	−18.40 to −10.58
Central Asia excl. Kazakhstan (median)	−0.37	−8.43	−6.10	−0.37 to +1.12	−12.01 to −4.33
Central Asia excl. Kazakhstan and Kyrgyzstan (value-weighted)	−3.01	−8.18	−10.36	−3.01 to −1.97	−13.30 to −5.78
Central Asia excl. Kazakhstan and Kyrgyzstan (median)	+2.06	+4.39	+5.12	+2.06 to +4.54	+0.02 to +9.87
Uzbekistan	−11.43	−27.97	−42.82	−18.31 to −11.15	−51.73 to −27.33
Central Asia + Russia (value-weighted)	−0.40	−3.54	−4.13	−0.40 to +0.40	−6.03 to −1.64
Central Asia + Russia (median)	+0.14	−1.21	−1.93	+0.14 to +0.95	−3.37 to +0.66
World	−1.46	+7.64	+7.62	−1.46 to −0.69	+5.50 to +8.75

Comparator (weighting)	2013→2019	2013→2024	2011–13 avg → 2023–24 avg	Range, end 2019	Range, end 2023–2024
Memo: Kazakhstan's imports from its other suppliers (Table 6.3)	+2.52	+14.11	+12.98		

Actual minus benchmark imports from China (§5.5). Positive: Kazakhstan's imports from China grew faster than the comparator. Central Asia excl. Kazakhstan = Kyrgyzstan, Uzbekistan, Tajikistan, Turkmenistan; the variants excluding Kyrgyzstan (a re-export hub) and Uzbekistan alone were added in the final validation. Ranges cover base years 2011–13 average, 2012, 2013, 2014. The median of Central Asia plus Russia coincides with Russia's growth factor. Source: Script 67, `robust_china_neighbour_grid.csv`.

Table 6.10c. Growth of China's exports to each partner (factor, China-reported)

Partner	2013, USD bn	2019, USD bn	2024, USD bn	2013→2019	2019→2024	2013→2024
Kazakhstan	12.55	12.73	27.95	×1.01	**×2.20**	×2.23
Russia	49.59	49.75	115.28	×1.00	×2.32	×2.32
Kyrgyzstan	5.08	6.28	19.90	×1.24	×3.17	×3.92
Uzbekistan	2.61	5.03	11.65	×1.93	×2.31	×4.46
Tajikistan	1.87	1.59	3.51	×0.85	×2.21	×1.88
Turkmenistan	1.14	0.43	1.03	×0.38	×2.39	×0.90
World	2,209	2,499	3,577	×1.13	×1.43	×1.62

Growth factors normalise for the very different sizes of the partners. Source: Script 67, `china_exports_growth_by_partner.csv`.

The comparisons have to be read period by period.

- **2019–2024, when the import rise happened.** China's exports grew by a factor of 2.2–3.2 to every one of Kazakhstan's five neighbours and by 2.20 to Kazakhstan, the lowest of the six (Table 6.10c). The rise in Kazakhstan's imports from China in this period was therefore not specific to Kazakhstan. This holds for each neighbour separately, so it does not depend on weighting or on including Russia or Kyrgyzstan.
- **2013–2019, the main construction period.** Kazakhstan's imports from China were flat (×1.01), slower than China's exports to the world (×1.13), Kyrgyzstan and Uzbekistan, and faster than to Tajikistan and Turkmenistan.
- **Whole period.** Over 2013–2024, whether Kazakhstan looks distinctive depends on the comparator: it is above Tajikistan and Turkmenistan, whose imports from China fell before 2019, and below Kyrgyzstan and Uzbekistan. In particular, against the median of Uzbekistan, Tajikistan and Turkmenistan, Kazakhstan's imports grew faster (+0.0 to +9.9 billion by 2023–2024). The claim, made in an earlier draft, that Kazakhstan's imports grew no faster "under every neighbour comparator" over the whole period was therefore too strong and has been withdrawn.

Three further qualifications apply.

1. *Comparators are not neutral.* After 2022, China's exports to Russia, and to Kyrgyzstan as a transit hub, were shaped by sanctions and re-routing. Kazakhstan was part of that pattern: its own exports of machinery, electrical equipment and vehicles to Russia rose from USD 0.33 billion in 2019 to USD 2.25 billion in 2023

(`russia_manufactures_check.csv`), although Comtrade does not record the origin of re-exported goods. The comparison shows that Kazakhstan's import rise belongs to a regional pattern; it does not establish the pattern's cause.

2. *Comparators are BRI participants, and infrastructure acts with lags.* Connectivity built in 2012–2019 (Table 6.13b) could have facilitated the post-2019 regional rise in China's exports. The comparison removes such region-wide effects together with other regional shocks, so it cannot rule them out. What it shows is that no *Kazakhstan-specific* mechanism is needed to account for the post-2019 import rise.
3. *Imports are measured with error.* Kazakhstan records far lower imports from China than China records exports (§4.5); the comparison uses China's data on both sides to avoid mixing reporters.

The import conclusion is therefore graded as follows: the post-2019 rise is **not specific to Kazakhstan** (robust: each neighbour separately); it does **not match** the M1 and M4 expectation of a Kazakhstan-specific expansion while BRI connectivity became operational (2013–2019 turnover and imports show no China-specific rise, §6.2); and whether region-wide BRI connectivity contributed to the post-2019 regional rise is **inconclusive**.

Table 6.11. Kazakhstan's imports from China by main chapter, China-reported (USD bn)

Chapters	2013	2019	2024
Apparel and footwear (HS 61, 62, 64)	4.24	3.28	7.67
Machinery (HS 84)	1.83	1.56	4.38
Vehicles (HS 87)	0.80	0.69	3.54
Electrical equipment (HS 85)	0.97	1.49	3.01
Furniture, toys, miscellaneous (HS 94–96)	0.46	1.21	2.05
Plastics and rubber (HS 39–40)	0.58	0.61	1.43
Iron and steel articles (HS 73)	0.77	0.60	0.83
All other chapters	2.90	3.30	5.04
Total	**12.55**	**12.73**	**27.95**

Source: UN Comtrade, China as reporter (Script 61, `import_composition_china_reported.csv`).

6.7 Inference Safeguards and Robustness

Table 6.12. Multiple-testing and randomisation checks

Test	Estimate	Unadjusted *p*	Holm-adjusted *p*	Randomisation check (300 random sets of 24 chapters)
Main: China × Post × metals and ores	0.329	< 0.001	< 0.001	n/a (12 chapters)
Main: China × Post × agri-food	2.456	< 0.001	< 0.001	Larger than all 300 draws ($p < 0.004$); median draw 0.39, 95th percentile 1.24
Main: China × Post × other	0.634	0.224	0.224	n/a
Trend change δ, agri-food	−0.368	0.020	0.059	Lower than 99.3% of draws (one-sided $p = 0.007$)
Trend change δ, metals and ores	−0.070	0.264	0.528	n/a
Trend change δ, other	0.107	0.437	0.528	n/a

Holm–Bonferroni within each family of three coefficients. Source: Script 66, `multiple_testing.csv`*,* `permutation_agrifood.csv`*.*

What survives. The China-specific rise in agri-food survives Holm correction and is larger than in any of 300 random sets of chapters of the same size, so it is unlikely to be an artefact of the choice of product group or of small-cluster inference. The metals result survives Holm correction in the main sample but not the exclusion of 2023–2024, so it is treated as a post-2022 result. No group shows a significant acceleration after 2013 (one-sided $p = 0.99$ for agri-food, 0.87 for metals and ores, 0.22 for other goods). The point estimates of δ are negative for agri-food and metals; for other goods it is positive but imprecise, which is absence of evidence rather than evidence either way.

What does not. The claim that the agri-food trend significantly *slowed* after 2013 is weaker than the other results: it fails the Holm correction at 5 per cent (adjusted $p = 0.059$), though the randomisation check ($p = 0.007$) supports it. The thesis therefore relies on "no acceleration", not on "slowdown".

Robustness to data source and missing data. With China's mirror data for the China leg, the export coefficients keep their signs and similar sizes (Table 6.7, row 2), and the 2020–2022 suppression does not enter because those years are excluded. In the decomposition, the China-specific share of the balance change is 92 per cent with either country's data (Table 6.3). The import-side conclusion for 2013–2019 is the only result that depends on the data source, and it is reported as such.

Independent re-estimation. Re-estimating the main model as a weighted binomial model of China's share in each chapter-year (§5.7) reproduces the PPML point estimates to four decimal places (metals and ores 0.3288, agri-food 2.4559, other 0.6340, energy −0.5280); clustered standard errors differ slightly because of small-sample adjustments (agri-food 0.220 against 0.213; metals and ores 0.077 against 0.074; other 0.538 against 0.522). Every main table was also regenerated from the raw files, and all values reproduced to within 0.001 (Appendix A).

What the tests could detect. The minimum change in the China-specific trend detectable with 80 per cent power at a one-sided 5 per cent level is about 0.39 log points per year for agri-food (SE 0.157) and 0.16 for metals and ores (SE 0.063). A BRI effect that raised the China orientation of these groups by less than this each year would not be detected. The non-significant trend changes for metals and other goods are therefore absence of evidence, not evidence of no change.

Extensive margin. In a linear probability model of exporting more than USD 1 million, the pooled China × Post effect is 0.013 (SE 0.040, $p = 0.75$): the number of chapters Kazakhstan exports to China did not rise relative to other markets. For agri-food alone the effect is 0.21 (SE 0.11, $p = 0.06$), from a base of 10 per cent (`extensive_margin_lpm.csv`). The extensive-margin prediction of M2 is not supported at HS-2 level; given the imprecision for agri-food, this is absence of evidence.

6.8 Do BRI Channels Match the Distinctive Components? (SQ4)

Table 6.13. Chinese official finance commitments to Kazakhstan by sector and period (USD bn, constant 2021)

Sector (AidData)	2000–2008	2009–2013	2014–2017	2018–2021	Total
Industry, mining, construction	15.88	19.57	6.38	2.81	44.64
Energy	2.75	7.21	0.36	0.13	10.46
Transport and storage	0.43	0.00	1.92	1.78	4.13
Other sectors	0.87	1.39	1.88	0.78	4.92

Sector (AidData)	2000–2008	2009–2013	2014–2017	2018–2021	Total
All sectors	**19.93**	**28.17**	**10.54**	**5.50**	**64.15**

Source: AidData GCDF 3.0, 155 projects recommended for aggregates (AidData, 2023). Author's calculations (Script 61, `aiddata_by_sector_period.csv`). No agriculture, forestry or fishing project is recorded.

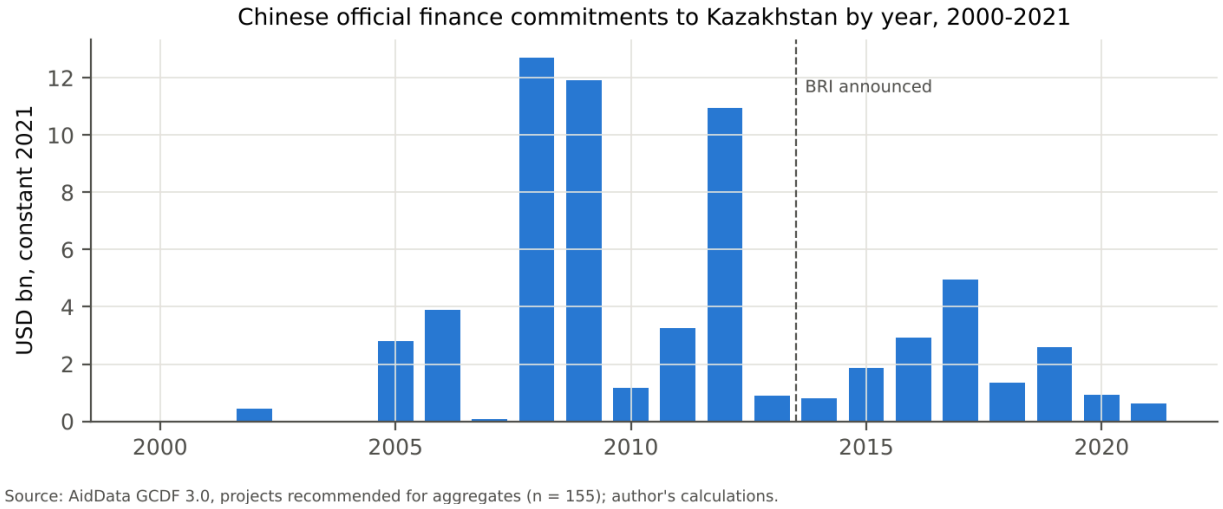


Figure 6.8. Chinese official finance commitments to Kazakhstan by year, 2000–2021

Recorded Chinese official finance to Kazakhstan was largest *before* the BRI: USD 48.1 billion (constant 2021 dollars) in 2000–2013 against USD 16.0 billion in 2014–2021. The largest pre-2013 commitments financed the Central Asia–China gas pipeline (2008, 2012), the acquisition of Kazakh oil companies (PetroKazakhstan 2005, Mangistaumunaigas 2009) and the Bozshakol (2009) and Aktogay (2011) copper mines (AidData records 61056 and 41941). After 2013 finance shifted towards transport: five China Eximbank road loans in 2017–2018 and a 2019 loan for border checkpoint modernisation. The Khorgos dry port began operating in 2015 (KTZ Express, 2025). Kazakhstan reported in 2023 that protocols covered 24 plant and livestock products (The Astana Times, 2023); the dates of individual protocols could not be verified.

Table 6.13b. Dated connectivity milestones relevant to Kazakhstan–China trade

Date	Milestone	Stage	Source	Verified?
December 2012	Altynkol–Khorgos rail border crossing opened (second Kazakhstan–China rail link)	Operational	Tengrinews (2012)	Yes (news report)
September 2013	Silk Road Economic Belt announced in Astana	Announcement	China Daily (2013)	Yes
2015	Khorgos Gateway dry port launched	Operational	KTZ Express (2025)	Yes (company source)
2016 (planned)	Completion of Kazakhstan's section of the Western Europe–Western China road	Planned completion	Kazinform (2015)	Plan only; actual completion date not verified
2017	Chinese state companies take 49% of Khorgos Gateway	Ownership	KTZ Express (2025)	Yes (company source)
2017–2018	China Eximbank loans for five road reconstruction projects	Financing commitment	AidData records 54989, 54990, 54991, 55016, 53554	Commitments verified; construction and completion dates not verified
2019	China Eximbank loan for border checkpoint modernisation	Financing commitment ("Pipeline: Commitment" status)	AidData record 70484	Commitment only

Date	Milestone	Stage	Source	Verified?
September 2025	Second track of the Dostyk–Moyynty railway inaugurated	Operational, after the sample	Fergana Agency (2025)	Yes (news report)

For most projects only announcement or financing dates could be verified; construction start and completion dates could not. The timing tests therefore use the dated operational milestones (2012, 2015) and the 2017 ownership and financing year, alongside the 2013 announcement (Table 6.9b). Because effects may arrive with lags, a change after 2019 is not by itself evidence against a BRI contribution, and the absence of acceleration is interpreted only for the windows tested.

Matching channels to components:

- **M3 (resource finance) and energy and minerals.** Resource finance was concentrated before 2013, and so was the rise in China's orientation of energy and metals (2005–2011). In the BRI period finance in these sectors fell, energy's China orientation fell, and metals' was flat until 2019. The China-specific gain in strategic minerals is positive in every specification, but its timing depends on the base year and does not coincide with the period of Chinese resource finance.
- **M1 and M2 (connectivity, market access) and agri-food.** Transport, border and market-access channels are post-2013, and agri-food is the one group whose China orientation kept rising after 2013 and which is most exposed to phytosanitary access and border speed. Recorded finance never went into agriculture, so if a BRI channel is involved it is access and logistics, not finance. But the reorientation began before these channels existed and did not accelerate when they appeared.
- **M1 and M4, imports.** A Kazakhstan-specific effect of connectivity or Chinese projects should have raised imports from China once infrastructure became operational (from 2012–2015 onwards). Instead, China's exports to Kazakhstan were flat in 2013–2019 and rose after 2019, as fast or faster to all five neighbours. Infrastructure completed by 2019 may have facilitated the later regional rise; these data cannot show whether it did.

6.9 Synthesis: Components, Mechanisms and What Is Established

Table 6.14. Components of the swing and their distinctiveness

Component	China-specific vs Kazakhstan's other partners (2013→2024; grid range to 2023–24)	Specific to Kazakhstan vs China's other neighbours?	Timing	Kind of evidence
Imports from China	+14.1 bn (+8.1 to +14.7)	No for 2019–2024 (lowest growth of six); comparator-dependent over 2013–2024	Mainly 2022–2024	Comparison-adjusted (descriptive)
Energy exports	−4.9 bn (−4.9 to +2.4; negative from every pre-2014 base)	Not assessed (comparators' exports dominated by oil and gas)	Concentrated 2013→2014	Comparison-adjusted; PPML point estimate without inference
Strategic minerals exports	+3.4 bn (+1.8 to +6.7)	Not assessed	Base-dependent	Comparison-adjusted; PPML significant only with 2023–24
Agri-food exports	+0.8 bn (+0.7 to +0.9)	Not assessed	Rising from 2008; no acceleration at 2012, 2013, 2015 or 2017	PPML, Holm, randomisation, trend tests
Other exports	+0.5 bn (+0.3 to +0.5)	Not assessed	Mainly after 2019	PPML (not significant)
Turnover (exports + imports)	−0.9 to +2.0 bn to 2019; +12.2 to +18.3 bn to 2023–24		Expansion after 2019	Comparison-adjusted

Imports enter the balance with a minus sign. Sources: Tables 6.3, 6.3b, 6.4, 6.7–6.10c, 6.12.

Mechanism scorecard. For each mechanism of §3.2: the expectation, its observable implication, the data needed, the evidence used here, what is established, what remains untested, and the verdict.

- **M1 Trade costs and connectivity.** *Expectation:* lower bilateral trade costs. *Implication:* China-specific expansion of bilateral turnover after operational milestones. *Data needed:* bilateral trade against benchmarks; ideally transport costs and times by route. *Evidence:* turnover comparison (§6.2), import comparisons (§6.6), milestones (Table 6.13b). *Established:* no China-specific turnover expansion to 2019; the post-2019 expansion was mainly imports and was shared by China's other neighbours. *Untested:* trade costs and transport times themselves; region-wide effects; lagged effects of 2012–2019 infrastructure on the post-2019 regional rise. *Verdict:* **inconsistent** with a Kazakhstan-specific expansion during 2013–2019; **inconclusive** about region-wide or lagged effects.
- **M2 Time-sensitive goods and market access.** *Expectation:* largest gains for food and non-traditional goods. *Implication:* China-specific rise in agri-food and other exports that starts or accelerates after a milestone; more chapters exported. *Data needed:* product-level exports to China and others; protocol dates; ideally HS-6. *Evidence:* PPML, event study, trend breaks at four years, extensive-margin model. *Established:* a large China-specific agri-food rise (robust to Holm and randomisation), starting before 2013, without acceleration at any milestone tested. *Untested:* product-by-protocol timing; HS-6 extensive margin. *Verdict:* **consistent in products, inconsistent with the predicted acceleration** at the milestones tested; extensive margin **inconclusive**.
- **M3 Resource finance.** *Expectation:* more China-oriented energy and mineral exports after Chinese finance. *Implication:* China-specific rises after finance and project completion. *Data needed:* product exports; project finance and completion dates. *Evidence:* decomposition, PPML, AidData. *Established:* energy fell specifically to China after 2013; strategic minerals rose in every specification; Chinese resource finance peaked before 2013. *Untested:* project completion and output dates. *Verdict:* **inconsistent** for energy; **inconclusive** for strategic minerals (positive but not timed with finance).
- **M4 Investment-linked imports.** *Expectation:* Chinese projects raise capital-goods imports. *Implication:* Kazakhstan-specific rise in machinery and equipment imports timed with projects. *Data needed:* project-level procurement; product-level imports from all suppliers. *Evidence:* import composition for three years (Table 6.11); regional comparison of totals. *Established:* machinery, electrical equipment and vehicles are 47 per cent of the 2019–2024 import rise; the rise was shared regionally. *Untested:* the link to specific projects. *Verdict:* **untested**.
- **M5 Transit.** *Expectation:* transit revenue. *Data needed:* transit volumes and service receipts. *Evidence:* none in this thesis. *Verdict:* **untested**.

What is established, what is interpretation, what is untested.

- *Established by the evidence (accounting and descriptive comparisons, robust across the specification grids):* the size and components of the swing; that most of it is China-specific relative to Kazakhstan's other partners (52–92 per cent); that turnover did not expand specifically with China before 2019; that the post-2019 import rise was shared by all five of China's other neighbours; that oil exports fell more to China than elsewhere; that strategic minerals offset part of the swing; that agri-food reoriented towards China (statistically robust) without acceleration at the milestones tested; that Chinese official finance peaked before 2013; that Kazakhstan's 2020–2022 partner data omit major chapters.

- *Interpretation:* that the evidence does not match the observable predictions of M1–M3 in the windows tested; that the post-2019 import rise is better described as a regional pattern than as a Kazakhstan–China development; that agri-food followed access and logistics rather than finance.
- *Untested:* any causal effect; M4 and M5; region-wide and lagged BRI effects; effects on investment, productivity and welfare; product-level timing at HS-6.

7. Discussion

7.1 The Answer to the Central Question

The central question asked whether the post-2013 change in Kazakhstan–China merchandise trade, summarised by the swing in the bilateral balance, is distinctive to the relationship and consistent with the BRI mechanisms observable in merchandise trade. The answer has three parts.

From Kazakhstan's side, the change is mostly China-specific. "China-specific" here has a precise meaning: the part of the change that would not have occurred had Kazakhstan's trade with China in each product grown at the rate of its trade in that product with the rest of the world. By that measure, between 52 and 92 per cent of the swing is China-specific, depending on base and end years, and 71 per cent with averaged endpoints (Table 6.3b). World commodity prices alone do not explain it. This is a statement about distinctiveness, not about causes.

Its largest component was shared by China's other neighbours. The largest China-specific component is the rise in imports from China after 2019. In 2019–2024 China's exports to each of Kazakhstan's five neighbours grew as fast as or faster than its exports to Kazakhstan (Table 6.10c). No Kazakhstan-specific mechanism is needed to account for it. Its causes, including post-2022 re-routing and any region-wide or lagged effect of BRI connectivity, cannot be separated with these data.

The merchandise-trade evidence does not show a change that both matches a BRI mechanism and starts or accelerates at a BRI milestone. Turnover did not expand specifically with China while connectivity projects were being opened (2013–2019). Energy exports fell specifically to China, contrary to the resource-security expectation. Strategic minerals offset part of the swing, but their China-specific gain is not timed with Chinese resource finance. Agri-food matches the market-access mechanism in products, but it began before 2013 and did not accelerate at any milestone tested (2012, 2013, 2015, 2017).

In the terms of §1.5: the BRI is supported as the **context (A)**; the agri-food reorientation is **consistent with a BRI mechanism in products and channel, but not in timing**, so it falls short of level **B**; and **no causal claim (C)** is made.

7.2 What These Results Do and Do Not Say About the BRI

- **Evidence inconsistent with specific predictions.** The absence of China-specific turnover growth in 2013–2019 (M1, Kazakhstan-specific expansion), the China-specific oil decline (M3, energy), and the absence of acceleration in agri-food at the milestones tested (M2, timing). These are the thesis's strongest negative findings, and each concerns a specific prediction in a specific window.

- **Absence of evidence.** The extensive margin, the “other goods” coefficients (positive but imprecise at every break year), and small trend changes. The minimum change in the China-specific trend detectable with 80 per cent power is about 0.39 log points per year for agri-food and 0.16 for metals and ores at the 2013 break (Tables 6.9, 6.9b). Effects of the size predicted by the ex-ante literature, a few per cent of trade (Baniya et al., 2020; Soyres, Mulabdic, Murray, et al., 2019), are well below what this design can detect.
- **No evidence of no effect.** Nothing here shows that the BRI had no economically meaningful effect on Kazakhstan. Transit services (M5), investment-linked imports (M4), productivity and welfare are not measured; region-wide and lagged effects cannot be excluded; and the outcome is merchandise trade only.

7.3 Relation to Existing Studies

The results sit alongside, rather than against, the literature that finds positive BRI trade effects.

Trade-cost and volume studies. De Soyres et al. (2019) and Baniya, Rocha and Ruta (2020) predict modest average reductions in trade costs and increases in trade, larger for time-sensitive goods. The thesis finds no China-specific expansion of Kazakhstan–China turnover in 2013–2019 relative to Kazakhstan’s other partners. This does not contradict those studies: their effects are averages of a few per cent across many economies, below what one pair’s annual data can detect, and they concern volume, not the balance. The agri-food result is the closest match to their prediction about time-sensitive goods, in composition if not in timing.

Region-wide effects. Krulikowski and Riker (2024) simulate that BRI membership raised China’s exports to Kazakhstan by about 29 per cent and Kazakhstan’s imports more than its exports, with the largest gains in Central Asia. That is consistent with two findings here, that most of the swing is China-specific from Kazakhstan’s side and that it was driven by imports, and with the finding that the import rise was shared by Kazakhstan’s neighbours, because their estimates are region-wide. The difference lies in timing and identification. Their estimates come from a model in which BRI membership (2015 for Kazakhstan) shifts trade costs; the observed data show China’s exports to Kazakhstan no higher in 2019 than in 2013 (after a fall in 2015–2016) and rising mainly in 2022–2024, the period of regional re-routing. The two approaches cannot be reconciled with the data available here: the neighbour comparison removes region-wide BRI effects together with other regional shocks, and their model attributes to membership whatever pair-level change follows it. This is the main reason the thesis classifies the regional and lagged contribution of the BRI to the import rise as inconclusive rather than absent.

Transit. Mau and Seuren (2023) show that China–Europe rail connections raised European exports to China. Kazakhstan’s benefit from such connections is primarily transit revenue, which is outside this thesis’s data. The thesis’s conclusions are therefore silent on the channel through which the BRI is best documented to have affected Central Asia.

Central Asian political economy. Area studies stress that the BRI is a broad label and that Central Asian trade has been shaped by transport costs, trade policy and the Eurasian Economic Union as well as by China (Laruelle, 2018; Pomfret, 2019). The finding that the largest component of the swing belongs to a regional pattern, and that the BRI-era channel with a measurable footprint is agricultural market access rather than finance, supports treating “the BRI” as a set of distinct channels with distinct timings rather than as a single treatment.

7.4 Why the BRI Remains the Right Frame

First, the question of whether the BRI changed Kazakhstan–China trade is the one asked in policy debate and in the registered topic, and it can only be answered by a design that compares the relationship with appropriate benchmarks. The thesis provides that design for merchandise trade.

Second, the change is China-specific from Kazakhstan's side. Had it been common to all of Kazakhstan's partners, the BRI would have been irrelevant to it and its place in the title decorative. Because it is China-specific, the BRI is the natural candidate, and because the BRI is both bilateral and regional, it had to be examined at both levels.

Third, the conclusions are about the BRI's observable mechanisms: which predictions the merchandise-trade evidence contradicts in the windows tested (Kazakhstan-specific expansion in 2013–2019; resource security for energy; acceleration of agri-food at milestones), where it is silent (region-wide and lagged effects, small average effects, investment-linked imports, transit), and what the change is instead (a China-specific change from Kazakhstan's side whose largest part, the post-2019 import rise, was shared by China's other neighbours).

7.5 The Swing Is Not a Loss: Interpreting the Balance

The swing is not evidence that Kazakhstan lost from trade with China. Of the USD 15.2 billion rise in imports from China in 2019–2024, USD 7.2 billion (47 per cent) was machinery, electrical equipment and vehicles, and USD 4.4 billion (29 per cent) apparel and footwear (Table 6.11). The first group can embody investment in Kazakhstan's capital stock, and both groups can lower prices for Kazakh firms and consumers; part of the first group also appears to have been re-exported to Russia (§6.6). The data cannot separate these uses. On the export side, the China-specific gains in strategic minerals and agri-food mean that Kazakhstan sold more of these products to China than its trade with other markets would imply. A bilateral deficit is consistent with all of these outcomes.

7.6 Strategic Minerals and the Balance

The registered topic asks about the role of strategic mineral exports. They **offset part of the swing**. Exports to China of ores, uranium compounds, ferroalloys, copper and zinc rose by USD 5.6 billion between 2013 and 2024, and their China-specific component is positive in every specification (USD 1.8–6.7 billion by 2023–2024; Table 6.4). Three qualifications apply:

1. In the 2013-base calculation, most of the China-specific gain came after 2019, mainly through ores, whose China share rose from 41 to 80 per cent. From earlier bases, a larger part (USD 1.7–2.4 billion) had occurred by 2019. The timing does not single out the BRI period.
2. The gain increased Kazakhstan's dependence on China as a buyer of ores and copper (§7.7).
3. The earlier version of this thesis found that the association between mineral exports and the balance weakened after 2013. The decomposition explains that pattern: the balance fell for reasons on other lines of the account (oil, then imports) while minerals rose, so a regression of the balance on minerals shows a weakening association without minerals having caused anything (Appendix B).

7.7 Dependence Is Two-Sided, but Mostly Asymmetric

Table 7.1. Two-sided dependence by chapter, 2024

HS	Chapter	China's share of KZ exports, %	KZ's share of China's imports, %
26	Ores	80.2	1.8
74	Copper	64.1	3.8
79	Zinc	50.3	28.1
28	Inorganic chemicals incl. uranium compounds	32.4	8.9
72	Iron and steel incl. ferroalloys	25.0	5.6
27	Energy	7.1	0.6
23	Food residues, fodder	67.3	4.2
15	Fats and oils	39.6	1.3
12	Oilseeds	41.0	0.2
10	Cereals	17.7	2.0

Source: UN Comtrade; Kazakhstan-reported exports and China-reported imports (United Nations Statistics Division, 2026a). Author's calculations (Script 64, `two_sided_dependence_2024.csv`*).*

For ores, copper and most agri-food chapters, China matters far more to Kazakhstan as a buyer than Kazakhstan matters to China as a supplier (Hirschman, 1945; Keohane & Nye, 1977). Zinc is closer to mutual dependence. Uranium is the main exception, and HS-2 data understate it because HS 28 mixes uranium with other chemicals: Kazakhstan supplied about two-thirds of China's natural uranium imports in 2023 (Pan, 2024) and 39 per cent of the world's mined uranium in 2024 (World Nuclear Association, 2026), and Chinese companies had to accept minority positions to secure access (Pan, 2024).

Dependence is not vulnerability. Ores and copper trade on world markets, and redirection is possible: China's share of Kazakhstan's refined copper exports fell from 65.4 to 49.1 per cent between January–November 2024 and the same period of 2025, while Türkiye's rose to 40.9 per cent (The Astana Times, 2026). The China-specific mineral gain raised Kazakhstan's sensitivity to Chinese demand; this thesis finds no evidence that it created leverage and does not assume that it did.

7.8 Alternative Explanations and What Remains Unexplained

- **China's own demand.** China-specific demand growth for vegetable oils and feed would produce the same agri-food component as BRI market access; the design cannot separate them (§5.6).
- **Post-2022 regional trade and lagged infrastructure effects.** The import rise and the metals shift both occur in 2023–2024. The neighbour comparison places the import rise in a regional pattern shared by all five neighbours. Post-2022 re-routing of trade with Russia is a plausible contributor, and connectivity built in 2012–2019 may have made that re-routing easier; the data cannot separate these, and the share of re-exports cannot be measured with Comtrade.
- **Oil.** The China-specific oil decline in 2014 is not explained here. Candidate explanations (routing of Kazakh crude, Chinese supply diversification, contract changes) could not be verified and are not asserted.
- **Pre-BRI Chinese investment.** Chinese finance for pipelines, oil companies and copper mines before 2013 coincides with the rise of China's share of energy and metals in 2005–2011. The coincidence is consistent with, but does not prove, a link.

7.9 Policy Implications

The recommendations follow only from findings that survived the robustness checks, and each states the finding it rests on.

1. **Treat the bilateral deficit as a diagnostic, not a verdict on the BRI** (finding: the largest component is a post-2019 import rise shared by all of China's neighbours, 47 per cent of it machinery, electrical equipment and vehicles). Government reporting on trade with China, for example by the Ministry of Trade and Integration, would be more informative if it separated capital goods, consumer goods and goods re-exported to third countries, rather than reporting the bilateral balance alone.
2. **Monitor the relationship against two benchmarks** (finding: whether a change looks China-specific depends on the comparison). Changes in trade with China are informative only when compared with Kazakhstan's trade with other partners and with China's trade with Kazakhstan's neighbours; a simple dashboard of the two comparisons, product group by product group, could be produced from public data each year.
3. **Monitor the concentration of strategic-mineral sales on China, without assuming leverage** (finding: China's share of ore exports rose from 41 to 80 per cent between 2019 and 2024, while the 2025 copper data show redirection is feasible). Buyer diversification for ores and copper is a commercial risk-management question for producers and for export-promotion bodies.
4. **Prioritise market access and border processing for agri-food** (finding: agri-food is the one group whose orientation towards China rose robustly, and no recorded Chinese finance went to agriculture). Extending phytosanitary protocols to further products and reducing border processing times are more directly connected to this channel than attracting Chinese loans.
5. **Reconcile partner-level trade statistics** (finding: Kazakhstan's 2020–2022 partner data omit major chapters, and its recorded imports from China are 1.5 to 2.5 times lower than China's records even in normal years). The Bureau of National Statistics could publish partner totals that include confidential chapters and a regular reconciliation with China's mirror data.

These implications do not depend on a causal estimate of the BRI's effect; they concern how the relationship is measured and where Kazakhstan's own policy levers are.

8. Conclusion

8.1 Answers to the Research Questions

The thesis asked whether the post-2013 change in Kazakhstan–China merchandise trade, summarised by the swing in the bilateral balance, is distinctive to the relationship, and whether it is consistent with the BRI mechanisms observable in merchandise trade, in particular through strategic mineral exports. Objective 1, deriving the BRI's mechanisms and their observable implications, is met in Chapters 2–3 (Tables 2.1 and 3.1). The answers to the sub-questions are:

- **SQ1 (what moved the swing, and is it China-specific?).** The balance moved from +USD 3.5 billion in 2013 to −USD 12.0 billion in 2024. The 2013–2019 phase was driven by energy exports, the 2019–2024 phase by imports. Relative to Kazakhstan's other partners, 52–92 per cent of the swing is China-specific, depending on base and end years, and 71 per

cent with averaged endpoints. Bilateral turnover did not expand specifically with China before 2019. *Objectives 2–3 achieved.*

- **SQ2 (which exports changed towards China, and when?).** Oil exports fell more to China than to other markets, mainly between 2013 and 2014. Strategic mineral exports rose specifically towards China in every specification, offsetting part of the swing. Agri-food exports reoriented strongly towards China, a result robust to multiple-testing correction and randomisation inference, but the reorientation began before 2013 and did not accelerate at any of the dated milestones tested. *Objective 4 achieved.*
- **SQ3 (was the import rise specific to Kazakhstan?).** No, for 2019–2024: China's exports grew as fast or faster to each of Kazakhstan's five neighbours. Over the whole period the comparison depends on the comparator. *Objective 5 achieved.*
- **SQ4 (do BRI channels match?).** Chinese official finance peaked before 2013 and went mainly to hydrocarbons and mining; post-2013 finance went to roads and border facilities; operational connectivity milestones date from 2012 and 2015. The timing and sectors of these channels do not match the components of the swing, except that agri-food, the group most exposed to market access, did reorient towards China. *Objective 6 achieved.*

Overall answer. The change in Kazakhstan–China merchandise trade is real and mostly China-specific from Kazakhstan's side. Its largest component, the post-2019 import rise, was shared by China's other neighbours. In the windows tested, the merchandise-trade evidence does not show a change that both matches a BRI mechanism and starts or accelerates at a BRI milestone. Strategic minerals cushioned the swing rather than causing it. These conclusions concern merchandise trade only; they do not show that the BRI had no effect on Kazakhstan, because transit, investment, productivity, welfare, region-wide and lagged effects are outside what the design can see.

Contribution. The thesis contributes a transparent, reproducible two-sided assessment for one country pair: it shows which parts of the change in Kazakhstan–China merchandise trade were distinctive, from whose side, and how each relates to the observable predictions of BRI mechanisms. It also documents a reporting gap in Kazakhstan's partner statistics that affects any analysis of this relationship.

8.2 Limitations

1. **No causal identification.** The comparisons measure distinctiveness; they are neither estimates nor bounds of BRI effects.
2. **Base-year sensitivity.** Shares and components depend on base and end years; 2013 was the peak year of oil exports to China. All key conclusions are reported as ranges over a grid (Table 6.3b, 6.4, 6.10b), and claims that held only for one base year have been removed.
3. **Comparators and timing.** The neighbour comparison is affected by post-2022 re-routing and sanctions, by re-export hubs and by the neighbours' own BRI participation; over 2013–2024 its result depends on the comparator, and only the 2019–2024 result holds for every neighbour separately. Construction and completion dates of most BRI projects could

not be verified, so timing tests use announcement, operational and financing milestones (Table 6.13b), and lagged effects cannot be excluded.

4. **Aggregation.** HS-2 chapters mix products (uranium with other chemicals). Imports from other partners were collected as totals, and China's exports to Kazakhstan by chapter only for 2013, 2019 and 2024.
5. **Power.** The tests can detect only large changes in trends (§6.7); small average effects of the kind predicted by the ex-ante literature are undetectable here.
6. **Energy is one chapter,** so no statistical inference is available for it.
7. **Missing years.** Kazakhstan's 2020–2022 product data are unusable for partner comparisons; Kazakhstan-to-World 2020 and 2022 and China-reported 2006 were not retrieved.
8. **Merchandise only.** Transit and other services, a main BRI channel for Kazakhstan, are outside the data.
9. **AidData** records official commitments, not disbursements or private investment.
10. **Exploratory elements.** Several analyses were added after earlier results were seen (§3.3); safeguards are described in §5.7.
11. **Outcome scope.** The balance is an accounting summary of merchandise trade, not a welfare measure; the thesis does not measure trade costs, investment, productivity or welfare.

8.3 Future Research

HS-6 data would allow testing the agri-food reorientation against the dates of individual phytosanitary protocols and measuring the extensive margin properly. Product-level import data for Kazakhstan and its neighbours would show which goods drove the post-2019 rise and how much passed on to Russia. Data on transit volumes and revenues would allow the BRI's main channel for Kazakhstan, which this thesis cannot observe, to be studied directly.

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Appendix A: Data and Code

Reproduction. From the repository root: `python Scripts/60_build_hs2_panel.py`, then Scripts 61, 62, 64, 65, 66, 67, 69 and 63 (figures), then `python Scripts/50_build_paper.py`, then `python Scripts/68_reconcile_manuscript.py`, which checks key figures in this manuscript against the output tables (34 of 34 found in the final version). All output tables were regenerated from the raw files during the final audit and reproduced the previous values to within 0.001. Python packages: pandas, numpy, statsmodels, pyfixest (0.60), matplotlib, openpyxl, scipy.

Table A.1. Scripts and outputs

Script	Purpose	Main outputs
60_build_hs2_panel.py	Parse raw HS-2 files, merge gap-fills, validate totals	`hs2_panel_long.csv`, `hs2_validation.csv`
61_composition_diversification.py	Composition, China shares, HHI, suppression check, balance decomposition, imports, AidData, Russia check, agri chapters	Tables 4.3, 6.1, 6.2, 6.5, 6.6, 6.11, 6.13
62_ppml_triple_difference.py	PPML triple difference, robustness, event study, pre-trend test, trend break, extensive margin	Tables 6.7–6.9, extensive margin (§6.7)
63_figures_reconstruction.py	Figures 6.1–6.8	`fig_r1`–`fig_r8`
64_two_sided_dependence.py	Two-sided dependence, 2024	Table 7.1
65_common_vs_china_specific.py	Shift-share split into common and China-specific components; strategic minerals; neighbour comparison	Tables 6.3, 6.4, 6.10, 6.10b
66_multiple_testing_permutation.py	Holm adjustment, one-sided acceleration tests, randomisation inference (300 draws, seed 20261031)	Table 6.12
67_robustness_baselines_comparators.py	Grids over base and end years; turnover; comparison with China's regional partners for both flows, value-weighted and median, with and without Russia and Kyrgyzstan; growth by partner	Tables 6.3b, 6.4 (ranges), 6.10b, 6.10c
68_reconcile_manuscript.py	Checks key manuscript figures against output tables	`reconciliation_report.csv`
69_timing_alternatives.py	Trend-break model with breaks at 2012, 2013, 2015, 2017 (exploratory)	Table 6.9b

Units and transformations. All trade values in current US dollars; PPML is estimated on values in USD million. Rest of world = Kazakhstan's exports to World minus exports to China (negative differences, which do not occur in the main sample, would be set to zero). Zero flows are kept. HS chapters are as reported by Comtrade in each year.

Data handling decisions.

1. Chapters missing from a truncated API response were re-queried and merged (gap-fill file); a chapter appearing twice keeps its first value.
2. Kazakhstan-to-World 2022 was rejected after inspection (duplicated values); 2020 could not be retrieved.
3. 2020–2022 are excluded from Kazakhstan-reported product comparisons because of suppression (§4.4).
4. Balances use China-reported data in both directions (§4.5).

Appendix B: Relation to the Earlier Specification

The earlier version of this thesis (archived as `Paper/archive/v2_minerals_trade_balance_dissertation.md`) regressed the annual bilateral balance (2000–2023, $n = 24$) and observed monthly balances (2010–2019, $n = 120$) on mineral exports, a post-2013 dummy and their interaction, with controls for oil prices and the exchange rate. Its main results were a positive pre-2013 association between mineral exports and the balance and a negative post-2013 interaction (annual $\beta = -2.49$; monthly $\beta = -0.91$, $p = 0.008$).

That specification was replaced for four reasons, each verified during the reconstruction:

1. **Identification.** A post-2013 dummy in a single country-pair time series cannot separate the BRI from the 2014–2016 commodity price collapse. The triple-difference design in Chapter 5 removes common shocks by comparing destinations.

2. **The omitted oil channel.** The monthly interaction disappeared once bilateral oil exports were controlled for, and Granger tests found that mineral exports do not lead the balance. Section 6.1 shows why: the 2013–2019 balance change is accounted for by oil.
3. **Data problems.** An "oil exports" column in the annual panel was a placeholder (55 per cent of total exports) and was removed; the 2020–2022 import data were shown to be incomplete (Table B.2).
4. **Inference.** A cross-partner difference-in-differences with six partners originally reported $p = 0.0002$ using standard errors unsuitable for a panel; corrected inference gave p between 0.02 and 0.38.

Table B.2. Kazakhstan-reported imports from China vs China-reported exports to Kazakhstan (USD bn)

Year	Kazakhstan-reported imports	China-reported exports	Ratio	Months with HS 84/85/87 import records
2013	8.36	12.55	1.50	12
2016	3.67	8.29	2.26	12
2019	6.57	12.73	1.94	12
2020	1.96	11.70	5.99	0
2021	2.42	13.76	5.67	0
2022	3.58	16.36	4.57	0
2023	16.77	24.70	1.47	12

Source: UN Comtrade monthly and annual data, both reporters (United Nations Statistics Division, 2026b). Author's calculations (Script 43, `import_break_evidence.csv`).

The scripts of the earlier analysis (Scripts 10–45) remain in the repository and run unchanged; their outputs are no longer cited in the main text.